Farm Household Misallocation

Jed Silver*

This Version: January 8, 2026
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Abstract

Agricultural markets often fail to allocate resources efficiently across farm households in developing countries. However, policymakers require knowledge of which markets fail and how the distortions they generate are correlated. Using data from rural Thailand, I characterize how distortions in land, labor, credit, and insurance markets each contribute to misallocation. I use moments in household consumption and production data to separately identify these distortions and develop a novel method using them to structurally estimate the production function. I find that the efficient allocation would increase aggregate productivity by 7.2% relative to the status quo, while only 4.0% (2.8%) gains could be achieved by eliminating financial (input) distortions in isolation. Positive interaction effects from addressing multiple distortions simultaneously account for the remaining 0.4% TFP gains. There also appear to be sharply decreasing marginal returns to addressing a single distortion in isolation, suggesting that policies overcoming inefficiencies in a broad set of markets would out perform attempts to eliminate a single one. Meanwhile, other common methods would produce larger estimates of misallocation and suggest that a financial market intervention would decrease aggregate productivity. Accounting for multiple correlated distortions is therefore crucial for measuring misallocation and designing policies to address it.

*Department of Environmental Economics and Management, Hebrew University of Jerusalem. Email: jedediah.silver@mail.huji.ac.il. I would like to thank Ethan Ligon, Jeremy Magruder, and Ted Miguel for their invaluable advice, guidance, and support. I would also like to thank Daniel Agness, Pierre Biscaye, Abdoulaye Cisse, Michelle Chen, Joel Ferguson, Lucy Hackett, Jonathan Levin, Peter Morrow, Martin Rotemberg, Elisabeth Sadoulet, Matthew Suandi, Elif Tasar, and audiences at the University of Oklahoma, Hebrew University, Reichman Univeristy, Bar-Ilan University and many conferences for helpful comments. All errors are my own.

1 Introduction

Agricultural productivity is up to 60 times higher in the richest than in the poorest countries (Gollin et al., 2014). A leading explanation is that markets in low-income countries do a poor job of allocating resources across farmers (Adamopoulos & Restuccia, 2014). At least qualitatively, this aligns with robust empirical evidence of incomplete credit, insurance, land, labor, and input markets in developing-country agriculture—often occurring simultaneously.[1] This paper asks: how much does each of these (potential) market failures contribute to overall misallocation and which should policymakers prioritize?

Answering these questions requires understanding not just the magnitudes of each distortion in isolation, but how they compound or offset others in equilibrium. Following the theory of the second best, the effects of reducing distortions in any single market are ambiguous and depend on the joint distribution of distortions in all markets (Lipsey & Lancaster, 1956). What determines a policy's effectiveness is not how much it reduces a particular distortion, but whether it moves producers closer to or further from the efficient allocation. For example, correcting distortions in land markets may have limited or even negative effects if the households that expand their landholdings are already inefficiently large due to preferential access to credit. Since considering a single market failure in isolation can lead to harmful policy recommendations, it is important to distinguish them empirically. Yet existing methods cannot separately identify the full set of distortions from production data alone (Hsieh & Klenow, 2009). Intuitively, data on inputs and outputs cannot reveal whether a household uses less of an input because it faces a higher price for it or cannot access credit or insurance. Moreover, dispersion in measured productivity could reflect measurement error rather than true distortions—a problem shown to severely inflate estimates of misallocation in agricultural settings (Gollin & Udry, 2021).

This paper separately identifies these distortions by exploiting the fact that, unlike typical firms, farm households are also consumers. Under imperfect markets, household consumption enters investment decisions and therefore contains information about how production is distorted (Benjamin, 1992). I leverage this fact to identify how credit constraints and uninsured risk distort productive choices distinctly from frictions in input markets. In particular, credit constraints are reflected in systematically lower consumption at planting than at harvest (reflecting the inability to smooth consumption across time by borrowing against future harvests). Meanwhile, uninsured production risk causes consumption at harvest to

[1] See Magruder (2018) and Suri and Udry (2022) for recent overviews.Bold et al. (2017), Breza et al. (2021), Caunedo and Kala (2021), Diop (2023), Goldstein and Udry (2008), Karlan et al. (2014), and Mobarak and Rosenzweig (2013) provide excellent examples of each of these market failures, respectively. Emerick et al. (2016) and Jones et al. (2022) are examples providing experimental evidence on how these market failures can compound each other.

depend on the realization of production shocks. On the other hand, input frictions function like a tax or subsidy and can be identified from dispersion in input *composition* across similar households.

More concretely, I jointly estimate the production function and these distortions using 196 months of panel data on inputs, output and consumption from Thai farm households. The key is that two sets of moment equations must hold. The first is simply the physical production function, relating *realized* output to inputs. The second is the demand function for each input, which under perfect markets would imply that households equate the *expected* marginal revenue product (identified from the first set of moment restrictions) of each input to a common factor price.[2] These are essentially asset pricing equations, in which the return structure of the assets (inputs) is governed by the physical production function. Financial frictions — namely credit constraints and uninsured risk — mean that marginal products in each state of the world are weighted by a stochastic discount factor, which can be estimated from consumption data.[3]. Meanwhile input frictions show up as systematic deviations from the law of one price after adjusting for the stochastic discount factor, and can be recovered using simple C-CAPM-like methods (Hansen & Singleton, 1982).

I then use estimates of the production function and distortions to calculate aggregate TFP under the observed allocation, and decompose misallocation into that due to factor market frictions, financial market frictions, and their interaction. Crucially, this is only possible when both input and financial distortions are well-specified. An important advantage of this decomposition is that it diagnoses the sources of misallocation while remaining agnostic towards the specific institutions that generate distortions. These distortions have many potential, possibly simultaneous, causes and conclusions may depend on which ones a model specifies. For example, recent empirical work has identified expropriation risk Goldstein and Udry (2008), incomplete contracting Burchardi et al. (2019), an explicit cap on landholdings Adamopoulos and Restuccia (2020), lack of titling Chen et al. (2022), land fragmentation Bryan et al. (2022), and others, as contributing to imperfect land markets. It would be impossible to capture all of these explicitly in a single model. Instead, my method allows me to diagnose how distortions in each market affect aggregate productivity without strong assumptions about their root causes.

With these estimates in hand, I turn to quantifying misallocation and its sources. I estimate overall misallocation — which I define as the TFP increase from the observed to efficient allocation — of 7.2%. This is even lower than estimates of total misallocation of

[2]The simplest example of this is calibrating Cobb-Douglas coefficients to observed revenue shares. However, these are not valid under imperfect markets because firms do not maximize expected profits and do not face common prices.

[3]Under a homothetic production function, this affects demand for all inputs symmetrically

19% in Shenoy (2017) from Thailand (albeit using different methodologies and data),[4] and substantially lower than estimates of 53% from China (Adamopoulos et al., 2022), 97% from Ethiopia (Chen et al., 2022), 259% from Malawi (Chen et al., 2023), and 286% from Uganda (Aragon et al., 2022).[5]

I then decompose these gains into the effects of eliminating either set of distortions in isolation and the interaction effect from eliminating them simultaneously.[6] I find that removing financial distortions while holding observed input wedges fixed would achieve 4.0–16.5% TFP gains relative to the observed allocation while removing input distortions alone would achieve 2.8–15.5% gains. Thus, TFP can be increased by a 0.4—12.3% (relative to baseline) by addressing both sets of distortions together. While the sign of these interaction effects is theoretically ambiguous, in the data it is positive because more financially constrained households are effectively subsidized in input markets.[7] Relatedly, I also find that there are sharply diminishing marginal returns to (uniformly) reducing distortions in a single market in isolation. Together these findings suggest that a multi-pronged policy approach addressing multiple frictions together could achieve larger gains from the same amount of resources than focusing intensively on a single market failure.

A further advantage of estimating the full set of distortions explicitly, rather than inferring unmodeled ones from a residual, is that it avoids conflating unmodeled frictions with measurement error. Both show up as dispersion in marginal revenue products, but only actual distortions represent misallocation. I show that methods which back out at least one distortion residually—as is common in the literature—would produce estimates over four times higher than mine in this setting, and would incorrectly suggest that removing financial frictions lowers aggregate productivity. While my approach requires additional modeling assumptions, this comparison illustrates how leaving distortions unmodeled can generate not just inflated but qualitatively misleading policy implications.

This paper's main contribution is developing a framework and estimation strategy to attribute misallocation to failures in distinct markets. Recent advances in the misallocation literature Carrillo et al. (2023), Hughes and Majerovitz (2023), and Sraer and Thesmar (2023) show how misallocation can nonparametrically be estimated from (quasi-)experimental

[4]In fact, given that I estimate returns to scale much closer to unity, one would expect my estimates of misallocation to be much larger, all else equal.

[5]This is more striking, given that the latter three papers only consider land misallocation.

[6]Note that this decomposition does not take into account the possible effects of relaxing one distortion on another — for example reducing land frictions through tenure reform could reduce credit constraints if it allows households to use land as collateral (Manysheva, 2021). However, modeling these potential effects would require taking strong stances on the types of institutions responsible for distortions and is beyond the scope of this paper.

[7]This is consistent with evidence that poorer households over-supply labor to their own farms because the shadow value of their time is lower Dillon et al. (2019) and Jones et al. (2022).

variation but are generally unable to trace misallocation to its different sources. There is also a growing literature applying quantitative misallocation models to microdata in agriculture (Adamopoulos & Restuccia, 2020; Adamopoulos et al., 2022, 2024; Aragon et al., 2022; Chari et al., 2021; Chen et al., 2017, 2022, 2023; Donovan, 2021; Gottlieb & Grobovšek, 2019; Manysheva, 2021; Shenoy, 2017). However, these papers typically model a single distortion in isolation or combine all distortions into a composite wedge. Notable exceptions are Manysheva (2021), who models the explicit dependence of credit constraints and land distortions through the collateral channel, and Shenoy (2017) who, also in Thailand, derives bounds for input and financial misallocation under assumptions on the joint distribution of distortions. In contrast, I estimate a more complete range of distortions that captures how they may compound or offset each other, which has important implications for policy.

Second, I contribute to the recent literature on how measurement error can inflate estimates of misallocation by using a model to separate between financial frictions and input mismeasurment. Rotemberg and White (2021) and Bils et al. (2021) find large upward biases due to measurement error in U.S. and Indian manufacturing. Meanwhile, Gollin and Udry (2021) argue that up to 70% of observed productivity dispersion in Ugandan and Tanzanian agriculture is due to measurement error and unobserved heterogeneity. This is supported by evidence of large and systematic measurement error in survey measures of agricultural land, labor, and output (Abay et al., 2019, 2021; Arthi et al., 2018; Desiere & Jolliffe, 2018).

Third, I contribute to the literature on production function estimation when input choices are distorted. This has been done in previous work to address adjustment costs (Asker et al., 2014), input price dispersion (De Loecker et al., 2016; Grieco et al., 2016), and markups (Asker et al., 2019; Cairncross et al., 2023; De Loecker & Warzynski, 2012), but not the types of distortions that farm households are likely to face, such as uninsured risk. My approach is to use a simple theory-consistent model of households' constrained optimal behavior to identify the production function given how input and financial frictions enter first-order conditions. Doing so ensures unobserved shocks' effects on input demands are subsumed by households' constrained-optimal choices of consumption and investment.

The rest of the paper is organized as follows: In Section 3, I present the theoretical framework and derive expressions for financial and input wedges at the household level, showing how they map to aggregate misallocation. Section 2 provides more information about the Thai data and context. Section 4 presents the estimation framework I develop and the results. Section 5 describes the decomposition of misallocation and Section 6 concludes.

2 Empirical Setting and Data

I use the Townsend Thai Monthly Survey, a panel dataset covering approximately 800 households in 16 villages across four provinces from 1998 to 2014. The survey collects detailed monthly information on agricultural production, household consumption, labor supply, and financial transactions, making it especially suited for jointly estimating production functions and financial frictions.

Of the 791 households in the sample, 568 engage in agriculture during the sample period. Over 68% of plots are cultivated with rice. In addition to crop production, households earn income from wages, livestock and aquaculture, and other businesses—the average agricultural household earns slightly less than half its income from crop cultivation.[8]

For the main analysis, I treat the township as the level of aggregation, since villages within townships are often quite integrated Kaboski and Townsend (2011) and Samphantharak and Townsend (2018). I focus on the sample of households cultivating annual crops during the main season, which I define as crops taking fewer than 8 months from planting to harvesting. I drop all plots that do not report using land or labor.

2.1 Agricultural production and inputs

Table C3 presents summary statistics for agricultural production. I focus on households cultivating rice, by farm the dominant crop in three of the four provinces, although I also show robustness to using all annual crops during the main season. I aggregate inputs to the farm-year level, since the model implies that shadow prices of inputs and consumption apply to all plots cultivated by a household at a given time. This yields a panel of 6,223 farm-season observations across 16 years.

I measure seven inputs: land, fertilizer, equipment (mainly tractors, power tillers, and pumps), seed, and labor at three stages of production (planting, weeding, and harvesting). Differentiating labor by production stage allows me to capture some of the seasonality in rural labor markets, where there may be tightness during planting and harvesting but slack at other times, without imposing a fully sequential production function.

Markets for all seven inputs exist, but vary considerably in depth. Land and labor markets are much more active in the Central region and appear quite thin in the Northeast. About 16% of farms rent land in a given season, but this masks substantial regional heterogeneity: nearly 40% of farms rent land in Chachoengsao while only 2.5% rent land in Sisaket. The

[8] While the model in Section 3 is a one-sector model, the estimation procedure I develop explicitly accounts for the fact that households obtain returns from other sources that may be correlated with those of crop cultivation. In fact, this is essential for matching the data.

average farm hires about 28% of its labor input, although more than two-thirds of farms hire some labor in a given season. Fertilizer, commercial seed, and mechanization are widespread and frequently acquired from outside the township. Table C4 plots coefficients of variation for input and output prices within the average township-year. There is very little price variation for fertilizer, seed, and rice, but substantial dispersion in wages, land rents, and equipment rental rates. This pattern supports my assumption that output is perfectly tradable within townships—so that the law of one price holds—while land, labor, and equipment markets may be subject to household-specific frictions or rations. Using the annual version of the dataset, Shenoy (2017) argues that input frictions reduce aggregate productivity by at least 6%.

2.2 Financial Markets

Households actively participate in both formal and informal finance, obtaining loans from government banks, credit schemes, neighbors, and informal moneylenders. However, most loans are small and only 5.7% are collateralized, suggesting that credit markets are far from complete. This is consistent with other research using the Townsend Data to study imperfect risk-sharing, borrowing constraints, and factor market imperfections. Kaboski and Townsend (2011, 2012) find that a microcredit expansion that occurred during the sample period partially relaxed binding credit constraints. Meanwhile, several papers suggest that kinship networks manage to share idiosyncratic risk fairly well Karaivanov and Townsend (2014), Kinnan and Townsend (2012), and Samphantharak and Townsend (2018) but far from perfectly, as idiosyncratic shocks propagate through labor supply and financial networks (Kinnan et al., 2024).

To estimate financial wedges, I require measures of the marginal utility of consumption at planting and harvest. The survey records monthly expenditures on 47 food and non-food goods, from which I estimate household-month marginal utilities using the procedure described in Section 4.1.2. I merge these estimates into the production panel to match the timing of input decisions and harvest realizations.

2.3 Why Thailand

Thailand during the sample period— covering the aftermath of the Asian financial crisis and its ascension to upper-middle-income status in 2011— offers a useful benchmark for understanding misallocation in developing-country agriculture. Unlike settings where the most dramatic estimates of agricultural misallocation have been obtained, Thailand lacks the severe institutional distortions such as explicit caps on landholdings (e.g. Adamopoulos

& Restuccia, 2020), de jure expropriation (Adamopoulos et al., 2024) or financial autarky in other settings. There is also substantial regional heterogeneity in the sample: The four provinces span Thailand's Central and Northeastern regions, which differ substantially in agricultural commercialization and market integration. The Central region (Chachoengsao and Lopburi) is closer to Bangkok, more commercialized, and better connected to output and input markets. The Northeastern region (Buriram and Sisaket) is poorer, more remote, and characterized by thinner markets and greater reliance on subsistence production. Nevertheless, Thai agriculture remains characterized throughout by rural smallholders rather than industrial farms. Therefore, Thai levels of misallocation may serve as a more reasonable benchmark for less developed countries than perfect markets or US agriculture.

2.4 Suggestive Evidence of Imperfect Markets in Thailand

Before imposing the structure of my model, I implement two canonical tests of complete markets. First, Townsend (1994) provides a test of full insurance: under complete risk-sharing, a regression of log consumption on log income with household and village-year fixed effects should yield a coefficient of zero, since idiosyncratic income shocks should be fully absorbed by the village risk-sharing network.[9] Second, Benjamin (1992) tests the joint null of complete markets by examining whether household composition—which affects preferences but not technology—predicts labor use. Under complete markets, production and consumption decisions are separable, so household demographics should be uncorrelated with input choices. Rejection is typically suggestive of labor market frictions causing households with larger labor endowments to supply more labor to their own farms, although it can in theory be associated with any market failure..

Table C1 presents the results. Column (1) reports the Townsend test: the coefficient on log income is 0.056 — in-line with recent estimates from India (Meghir et al., 2022) and highly significant. Columns (2) and (3) report the Benjamin test using household size and detailed age-sex composition, respectively; both reject separability, the latter at the 5% level.

While these regression coefficients do not have structural interpretations, their joint distribution is informative about how frictions may interact. I run both tests separately for 64 village $\times$ 4-year blocks and plot the coefficients against each other in Figure C1. If financial and input market frictions were independent across villages, we would expect no systematic relationship. Instead, the coefficients are negatively correlated: villages where consumption is more sensitive to income shocks (indicating worse insurance) also exhibit stronger dependence of labor use on household composition (indicating more distorted labor markets).

[9] Ligon (2025) illustrates the importance of using a measure of the marginal utility of consumption instead of log consumption if preferences are non-homothetic and shocks are covariate.

This correlation is consistent with financial and input frictions compounding each other—households that cannot smooth consumption may also be unable to hire labor at market wages, forcing them to rely on family labor regardless of its opportunity cost. However, these reduced-form tests cannot quantify the contribution of each friction to aggregate misallocation or determine whether the frictions offset or reinforce each other in equilibrium. Doing so requires the structural approach I develop in the following sections.

3 Model

I apply the workhorse farm-household model (Singh et al., 1986) to characterize how frictions in financial and input markets generate distinct wedges in households' input demands. The goal is not to microfound the specific institutions that generate these wedges, but to derive model-consistent measures of distortions that can be estimated from data on production and consumption. This agnosticism toward root causes is deliberate: distortions in any given market may arise from many sources simultaneously, and taking strong stances on which ones generate observed frictions would require additional assumptions that the data cannot discipline. For example, financial frictions could stem from limited commitment in informal insurance networks (Ligon et al., 2002), moral hazard in credit markets (Karlan & Zinman, 2009), or simply the absence of formal financial institutions (Karaivanov & Townsend, 2014). Likewise, land market distortions could reflect expropriation risk (Goldstein & Udry, 2008), incomplete contracting (Burchardi et al., 2019), explicit caps on holdings (Adamopoulos & Restuccia, 2020), or lack of titling (Chen et al., 2022). My approach captures the aggregate effect of these frictions on input allocation without requiring me to specify which combination generates them. The cost of this generality is that I cannot model how relaxing one distortion might affect another—for instance, how land titling might ease credit constraints through the collateral channel (Manysheva, 2021). I return to this limitation when interpreting the decomposition results.

3.1 Setup

The environment is standard. Households i have time-separable preferences over a vector of consumption goods (c) and leisure (l), with per-period utility $u(c_{it}, l_{it})$ and maximize expected lifetime utility. They produce a single homogeneous agricultural good with production given by

$$Y_{it+1} = F(Q_{it}, \psi_{it+1}). \tag{1}$$

where Q_{it} is a vector of K inputs applied by i at time t, and ϕ_{it+1} is a shock realized at

time $t+1$, prior to harvesting output Y_{it+1}. As is standard, I assume that $F_k > 0, F_\varphi > 0$,and $F_{kk} < 0$ for each input k. Households may hold a portfolio of assets with (possibly) stochastic returns. Each period, households choose consumption, input quantities for the following period's harvest, and asset holdings, subject to a budget constraint. They may also face binding credit constraints that prevent borrowing against future output, incomplete insurance that leaves consumption exposed to production shocks, and frictions in input markets that cause effective prices to deviate from market prices. The full dynamic program is in Section A.

A first set of FOCs requires that households set the marginal utility of consumption from each good j equal to its price p_{jt} times the Lagrange multiplier associated with the budget constraint, λ_{it}, which can be thought of as the marginal utility of expenditure.

$$u_j(c_{it}, l_i t) = p_{jt}\lambda_{it} \tag{2}$$

Note that the budget constraint must hold in each state of the world, and thus λ_{it} is a random variable as of periods before t (absent full insurance). For simplicity, I assume that the law of one price holds in goods markets.[10]

The frictions household may face (directly) enter the first-order condition for input k as two wedges, τ_{ikt} and Λ_{ikt}:

$$\bar{w}_{vkt}\tau_{ikt} = \delta\mathbb{E}_t\left[F_k(q_{it}, \varphi_{it+1})\right]\Lambda_{it}, \quad \Lambda_{ikt} \equiv \frac{\mathbb{E}_t\left[\lambda_{it+1}F_k(q_{it}, \varphi_{it+1})\right]}{\lambda_{it}\mathbb{E}_t[F_k(q_{it}, \varphi_{it+1})]} \tag{3}$$

where $\bar{w}_{vkt}$ is the market price of input k and F_k is its marginal product (under a general well-behaved production function).

Input wedges τ_{ikt} capture frictions that make the effective price of input k differ from the market price for household i. These include explicit taxes or subsidies, transaction costs, and binding quantity constraints. When $\tau_{ikt} > 1$ ($\tau_{ikt} < 1$) the household acts as if it faces a tax (subsidy) relative to the market price. Meanwhile, Λ_{ikt} captures how credit constraints and uninsured risk distort the intertemporal tradeoff between expenditure at planting and consumption at harvest. This can be decomposed into two terms. The first, $\mathbb{E}_t[\lambda_{it+1}]/\lambda_{it}$, reflects credit constraints: when households cannot borrow against future harvests, marginal utility at planting exceeds expected marginal utility at harvest. The second reflects uninsured risk, in which consumption at harvest depends on production shocks — expressed as the deviation of the realized marginal product of k from its expectation.

This expression holds for any well-behaved production function and utility function — financial frictions operate through the stochastic discount factor while input frictions operate

[10] This is consistent with the minimal cross-sectional variation in food prices within a township in the data.

through effective prices—does not depend on functional form. However, for the estimation, I will assume production takes the familiar Cobb-Douglas form with decreasing returns to scale $\gamma \equiv \sum_k \alpha_k < 1$.[11]

$$Y_{it+1} = \Psi_{it+1} \prod_k Q_{ikt}^{\alpha_k} \tag{4}$$

Without loss of generality, I assume $\Psi_{it+1} = e^{a_{it}+\phi_{it+1}}$, where a_{it} is anticipated productivity and ϕ_{it+1} is an unanticipated shock (from a known distribution) that cannot be predicted with time t information. I assume further $\mathrm{\$a_{it}}$ follows a first order Markov process — $a_{it} = h(a_{it-1}) + \xi_{it}$ — in which $h(\cdot)$ is a general function capturing the anticipated evolution of productivity and ξ_{it} is the unpredictable "innovation" to productivity, orthogonal to household's prior information sets by construction. I assume decreasing returns to scale with $\gamma \equiv \sum_k \alpha_k < 1$.[12]

Under the Cobb-Douglas assumption. I can rewrite (22) to obtain the demand function for each input k. In an appendix, I also extend this framework to a nonhomothetic production function featuring differentially risky inputs, following Just and Pope (1978, 1979).

To ease notation, I write levels of anticipated productivity (as of time t) as $A_{it} \equiv \exp(a_{it})\mathbb{E}[\exp(\phi_{it+1})]$ and the unanticipated component as $\varphi \equiv \frac{\exp(\phi_{it+1})}{\mathbb{E}[\exp(\phi_{it+1})]}$. This ensures that $\mathbb{E}[\varphi_{it+1}] = 1$ by construction.[13]

In this case, the FOC for input demands can be easily arranged as an input expenditure function.

$$\begin{aligned} q_{ikt}\bar{w}_{vkt}\tau_{ikt} &= \delta\alpha_k \frac{\mathbb{E}_t[\lambda_{it+1}Y_{it+1}]}{\lambda_{it}} \\ &= \delta\alpha_k \mathbb{E}_t[Y_{it+1}]\Lambda_{it}, \quad \Lambda_{it} \equiv \frac{\mathbb{E}_t[\lambda_{it+1}\varphi_{it+1}]}{\lambda_{it}} \end{aligned} \tag{5}$$

In (5) the financial wedge Λ_{it} no longer may differ across inputs (it is no longer indexed by k) and reduces to the production shock φ, weighted by the intertemporal marginal rates of substitution associated with different values of φ. In particular, if there is full insurance, then $cov_t(\lambda_{it+1}, \varphi_{it+1}) = 0$ and $\Lambda_{it} = \frac{\mathbb{E}_t[\lambda_{it+1}]\mathbb{E}_t[\varphi_{it+1}]}{\lambda_{it}} = \frac{\mathbb{E}_t[\lambda_{it+1}]}{\lambda_{it}}$ (since $\mathbb{E}_t[\varphi_{it+1}] = 1$ by construction). Under perfect credit markets, this further reduces to $1/\delta$ divided by a common interest rate R_t and there is no misallocation.[14] Meanwhile, dividing (5) for any two inputs makes clear

[11] Since there are no fixed factors, constant returns to scale would imply the efficient allocation is degenerate, with only the produce with the highest A_{it} producing. However, I do not impose the constraint $\gamma < 1$ during the estimation.)

[12] If $\gamma \geq 1$, then the efficient allocation is degenerate with only the most productive producer producing.

[13] $\mathbb{E}[\exp(\phi_{it+1})]$ enters the anticipated productivity term because the distribution of ϕ_{it+1} is known in advance.

[14] As will become clear in Section 3.2, this R is a constant that falls out of the expression for misallocation

that the difference in input *composition* only depends on τs, as Λ_{it} cancels out. This is true for any homothetic production function.

3.2 Equilibrium

I now show how this model of farm-household distortions maps to aggregate misallocation.

Let $\eta \equiv \frac{1}{1-\gamma}$, which is a nonlinear transformation of returns to scale that approaches ∞ as production approaches CRS. In what follows, I drop time subscripts to ease notation. A decentralized allocation yields the following expression for the share of factor k in a given location allocated to household i. [15]

$$\omega_{ik} \equiv \frac{\frac{1}{\tau_{ik}} \left(A_i \Lambda_i \prod_l \tau_{il}^{-\alpha_l}\right)^{\eta}}{\sum_h \frac{1}{\tau_{hk}} \left(A_h \Lambda_h \prod_l \tau_{hl}^{-\alpha_l}\right)^{\eta}} \tag{6}$$

(6) is obtained by aggregating household first-order conditions (28) and implies that any allocation can be defined as a function of technology α, household TFP A, and distortions Λ and τ. This also implies that when there are no distortions (i.e. $\Lambda_i = \tau_{ik} = 1$ for all inputs and households), the optimal allocation is

$$\omega_i^* \equiv \frac{A_i^{\eta}}{\sum_h A_h^{\eta}} \tag{7}$$

In this case, each input is allocated proportionally to household TFP, transformed by returns to scale.[16]

(6) shows how financial and input distortions jointly determine the allocation. A household with a low Λ_i (severe financial constraints) produces at an inefficiently small scale, but this can be either amplified or offset by input wedges If it faces $\tau_{ik} > 1$ for some input k, (effectively taxed in the market for that input), both distortions push in the same direction and compound each other. But if $\tau_{ik} < 1$, then the effective subsidy partially offsets the financial friction. This is why the effect of reducing a single distortion in isolation is ambiguous — what matters for allocative efficiency is not whether it shrinks the distortions towards 1 but whether it pushes ω_{ik} closer to — or farther from — ω_i^*.

in equilibrium.

[15] Note that anything constant across households within the same location — such as the discount factor, market price of inputs, their aggregate supply and the market interest rate — cancel out of (6) .

[16] This is a standard result in the misallocation literature.

4 Estimation Framework

I now describe how to estimate the four main objects that determine misallocation: Technology parameters (α), household anticipate TFP (A), financial wedges (Λ), and input wedges (τ). I proceed in two steps, first estimating α and (a parameterized version) of τ using GMM and then predicting A and Λ by isolating the ex ante predictable components of realized TFP and consumption.

With estimates of α in hand, I can construct the ex post realization of TFP and its covariance with consumption. But recall from Section 3 that A and Λ are households' ex ante *expectations* of these objects. In a second step, I therefore project the realized values of TFP and $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ onto the rich set of households' financial variables to isolate their anticipated components from realizations of ex post shocks, which cannot be considered misallocation (Gollin & Udry, 2021). The predicted values are my estimates of A and Λ, which I use in the decomposition exercise in Section 5.

4.1 Parameter Estimation (α and τ)

The model implies that both the physical production function and the first-order conditions for input demands must hold. The production function relates realized (ex post) output to inputs while first-order conditions relate input expenditure to (ex ante) expectations over future consumption and marginal products. I now estimate the technology parameters α and a parameterized version of input wedges τ via GMM using these as moment conditions. While I estimate them jointly, I first describe each of the moment conditions in turn.

4.1.1 Production Function

I begin with the production function, which identifies the output elasticities α. Taking logs (denoted by lowercase letters) of the Cobb-Douglas specification (4) and first-differencing yields:

$$\Delta y_{it} = \sum_k \alpha_k \Delta q_{ikt} + h(a_{it-1}) - a_{it-1} + \xi_{it} + \Delta\phi_{it+1} \tag{8}$$

I propose using lagged inputs q_{ikt-1} as instruments for Δq_{ikt}, following the Anderson and Hsiao (1981, 1982) approach used by Shenoy (2017) and Manysheva (2021). However, the residual now contains four terms, $h(a_{it-1}) - a_{it-1} + xi_{it} + \Delta\phi_{it+1}$. The q_{ikt-1} instruments are orthogonal to ξ_{it} by the Markov assumption and to ϕ_{it+1} and ϕ_{it}, which are unanticipated. However, the anticipated change in productivity between $t-1$ and t, $h(a_{it-1}) - a_{it-1}$, needs to be proxied for.[17] I proxy for this anticipated component $h(a_{it-1}) - a_{it-1}$ with $g(z_{it-1})$,

[17] In standard proxy variable approaches in IO (e.g. Gandhi et al., 2020), the assumption of perfect markets

where $z_{it-1} \subseteq \mathcal{I}_{it}$ is a function of households' consumption, investment and other financial variables in households' $t-1$ information sets. This is a valid proxy if $g(z_{it-1})$ is invertible, which corresponds to households systematically adjusting their consumption and investment in anticipation of future productivity shocks.[18]

Expressing this as a moment condition (which, again, is estimated jointly with the input demands):

$$\mathbb{E}\left[q_{ikt-1} \otimes \left(\Delta y_{it} - \sum_k \alpha_k \Delta q_{ikt} - g(z_{it-1})\right)\right] = 0 \tag{9}$$

The identification assumption is that lagged inputs q_{ikt-1} are orthogonal to the innovation shock ξ_{it}, the difference in unanticipated productivity shocks $\phi_{it+1} - \phi_{it}$ and the error from approximating $h(a_{it-1}) - a_{it-1}$ with $g(z_{it-1})$.

This approach has an important advantage in settings with market failures. Standard proxy-variable methods following Olley and Pakes (1996) assume that a flexible input can be inverted to recover unobserved productivity, which requires TFP to be the only unobservable affecting input demand. This assumption fails when wedges also enter input demands, as they do here.[19] However, in such cases, households' consumption problems become inseparable from their production problem and information about anticipated productivity can be inferred from changes to their consumption and investment patterns.

4.1.2 Euler Equation

The other set of moment conditions is the Euler-like equation for input investment, which identifies the τs and Λ under rational expectations. Under complete markets, households would simply set the (expected) expenditure shares of output for each input k to α_k, allowing us to simply calibrate α_k as observed expenditure shares. However, in general, output in each state of the world is weighted by a stochastic discount factor and the law of one price for inputs may not hold. These deviations from the perfect market benchmark are captured through the Λ and the τ, which we estimate to make the first order condition for input demands hold.

Estimating marginal utilities (λ) The financial wedge Λ depends on marginal utilities of consumption at planting (λ_t) and at harvest (λ_{t+1}). I estimate these λs from disaggregated expenditure data on 47 food and non-durable goods observed monthly over the 16-year

allows one to invert the demand function of a flexible input to proxy for the anticipated TFP shock.

[18] This is likely especially relevant when borrowing constraints bind, requiring households to substitute away from consumption and other investments to finance the desired increase in input investment.

[19] As Shenoy (2021) notes, the market failures that cause misallocation break the "functional dependence" of inputs on anticipated TFP shocks, which would otherwise make proxy-variable methods unidentified.

panel, using the Constant Frisch Elasticity demand system proposed by Ligon (2020). This procedure, which identifies λ up to an exponent ζ that I estimate jointly with the input demand parameters,[20] relates log expenditure on good j to prices, household characteristics, and the marginal utility of expenditure:

$$\mathbb{E}[\log x_{ijt} - \alpha_j(p) + \delta(z) + \beta_j \log \lambda_{it}] = 0 \tag{10}$$

where p is the vector of prices, z is a vector of household characteristics, and β_j is the elasticity of expenditure on good j with respect to the marginal utility of expenditure λ_{it}. This structure accommodates highly nonlinear Engel curves and nests many commonly used demand systems. I estimate λ using the full 196-month panel, allowing demands to vary with household composition as measured by the counts of members in different age-sex bins. Further details of the estimation procedure are in Appendix D.

Estimating τs With estimates of λ and α and observations of input quantities q and market prices $\bar{w}$, all that is left to estimate from (5) are τ, δ, and the nuisance parameter ζ. For δ, I convert Kaboski and Townsend (2011)'s estimate of an annual discount factor of 0.926 — also using the Townsend Thai Data — to reflect the number of months between the application of input k and harvest.[21]

Now I turn to estimating the τs (and ζ) via GMM. In practice τ_{ikt} needs to parameterized. This is because there are as many τ_{ikt}s as there are observations in the data and the residual from (11) would pick up any ex-post shocks to λ_{it+1} and Y_{it+1} as well as measurement error. In order to ensure that the estimates of τ_{ikt} pick up only variation that looks like an input distortion, I parameterize it as a log linear function of lagged expenditure share of input k and household composition (i.e. the counts of male and female adults and children). The former reflects that fact that input distortions are likely to be serially correlated while the later reflects the well-established correlation between labor intensity and household size (LaFave & Thomas, 2016).

For instruments Z_{it}, I include lags of λ_{it} from the three months before input k is applied and the lagged value of the input from the prior season's production, along with a constant. The corresponding population moment condition becomes:

$$\mathbb{E}[Z_{ikt}(\bar{w}_{kvt} Q_{ikt} \tau_k(Z_{ikt}) - \alpha_k \delta \left(\frac{\lambda_{it+1}}{\lambda_{it}}\right)^{\zeta} Y_{it+1})] = 0 \tag{11}$$

[20] ζ is what Frisch (1959) calls "money flexibility" and will in practice also absorb sampling variation in the estimates of λ.

[21] I also show robustness to assuming an annual discount factor of 0.95.

where

$$\tau_k(Z_{ikt}) = \rho_{0k} + \rho_{1k}(Q_{ikt-1}/\sum_l Q_{ilt-1}) + \rho_{2k}\text{AdultMales}_{it} + \rho_{3k}\text{AdultFemales}_{it} \\ +\rho_{4k}\text{ChildMales}_{it} + \rho_{5k}\text{ChildFemales} + \nu_{ikt}. \tag{12}$$

Why does this work? (11) is an *un*-conditional moment condition (note the absence of the t subscript on the expectation operator indicating conditioning on the household's information set). Meanwhile, the expectation operator in the first order condition (5) represents households' *subjective* expectations over future production shocks. While we can't observe these without full knowledge of households' information and beliefs, if expectations are rational, then we know that they are correct — and thus equal to what we'd expected to observe in the data — *on average*.[22]

As a robustness check, I replace the estimated λs in (11) with monthly consumption values; under CRRA preferences this gives ζ the interpretation of the coefficient of relative risk aversion. Both approaches yield similar results (Appendix X).

4.1.3 GMM Implementation and Results

I use a linear function $g(z_{it-1}) \equiv z'_{it-1}\pi$ to proxy for the anticipated productivity shock $h(a_{it-1})$, using variables from households' $t-1$ balance sheets. To avoid overfitting, I apply a multiple post lasso following Belloni et al. (2014) to select only those variables that predict Δy_{it} or any of the Δq_{it}s, the list of which is in Section E. This process winds up selecting 18 covariates.

The only calibrated parameter is the discount factor δ. I set the annual discount factor equal to Kaboski and Townsend's 2011 estimate of 0.926. However, unlike in the model, in which a period roughly corresponds to a season, different inputs are applied with varying time until the harvest. I therefore convert the annual discount factor to reflect the number of months remaining until the actual harvest.[23]

There are 50 remaining parameters to be estimated from 92 moment conditions. These include seven moment conditions to identify the seven αs,and 12 instruments (constant, lagged input expenditure and share, four household composition variables and five lags of λ)

[22] A standard concern with GMM estimators under rational expectations is that they require T to be large for aggregate shocks to average out — the sample mean of (11) is a random variable that is zero in expectation but is not necessarily equal to zero in a given year. With small T, realized aggregate shocks may have non-zero correlation with the instruments, biasing the estimates. While 16 years of data is modest by macroeconomic standards, I show via Monte Carlo simulations in Appendix X that the resulting finite-sample bias is likely negligible in my setting.

[23] Note that this assumes that households either know when they will harvest their crop, or that any uncertainty over how many months remain until harvest is uncorrelated with future consumption.

to identify six ρs for each of the 7 inputs plus a common ζ.

Table 1 reports the estimates and standard errors for the αs and their sum γ, in Column (1). The coefficients each take reasonable values for agricultural production functions, as does returns to scale $\hat{\gamma} = 0.79$, which implies an η of 4.79. The estimates of the ρ are not particularly notable, except for the negative and significant coefficients on lagged input intensity for all inputs except fertilizer, which suggest persistence in the τs—households that used an input less intensively in previous seasons behave as if they currently face a higher price for it. I fail to reject the overidentifying restrictions of the model, with a J statistic of 45.88 ($p = 0.28$ (42 degrees of freedom)).

The remaining columns of the table also report the estimates of ρ that are used to construct τ for each input, as well as the "money flexibility" parameter ζ. Note that $\hat{\rho}$ is only used to construct the τs used in the main analysis, while ζ is discarded. Histograms of each of the estimated $\hat{\tau}_k$ as well as the Cobb-Douglas price index $\prod_k \tau_{ikt}^{alpha_k}$ are plotted in Figure C7. The most dispersion is observed for planting and harvest labor, while the least is observed for fertilizer. This aligns with the observed within tambon-month price dispersion in each of these inputs (Table C4), in which the law of one price appears to hold for fertilizer but not for other inputs. Moreover, while the average τ_k with a tambon-year does not affect the main counterfactual results as it divides out of (6), the average values below 1 for fertilizer and weeding labor suggest that they are effectively subsidized, while other inputs are effectively taxed..This again conforms to conventional wisdom, in which fertilizer is heavily subsidized by the Thai government[24] and labor markets are slack during the midseason (weeding) period and tight otherwise. #+begin$_{\text{export}}$ latex

#+end$_{\text{export}}$ latex

4.2 Productivity and Financial Wedges (A and Λ)

The second step of the estimation procedure is to decompose realized TFP into anticipated (A) and unanticipated (ϕ) components, and to construct financial wedges Λ. Both require separating what households could predict ex ante from ex post realizations.

[24]Note that the fertilizer subsidy in Thailand is applied at the national level and therefore does not contribute to input wedges in the model as it does not distort the *relative* prices faced by farmers within a tambon-year.

Table 1: GMM Parameter Estimates

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.228***	-0.197	-0.465***	-0.023	0.064	0.017	-0.013
	(0.024)	(0.174)	(0.065)	(0.029)	(0.037)	(0.037)	(0.031)
Fert.	0.044***	-0.836**	-0.042	-0.070**	0.096**	0.001	-0.010
	(0.011)	(0.287)	(0.050)	(0.030)	(0.039)	(0.036)	(0.033)
Harv. Labor	0.127***	-1.583***	-0.356**	0.332**	0.385***	-0.045	-0.004
	(0.017)	(0.322)	(0.136)	(0.102)	(0.110)	(0.053)	(0.045)
Land	0.185***	-0.057	-0.282***	0.002	0.075*	0.005	-0.019
	(0.020)	(0.169)	(0.068)	(0.027)	(0.034)	(0.034)	(0.026)
Plant. Labor	0.065***	-0.271	-0.365***	-0.018	0.051	0.020	-0.026
	(0.012)	(0.313)	(0.056)	(0.033)	(0.037)	(0.039)	(0.032)
Seed	0.121***	0.517*	-0.241***	-0.041	0.043	0.021	-0.029
	(0.016)	(0.252)	(0.050)	(0.027)	(0.036)	(0.033)	(0.027)
Weed. Labor	0.021**	-1.894***	-0.351***	-0.020	0.159**	-0.011	0.043
	(0.007)	(0.383)	(0.046)	(0.042)	(0.057)	(0.058)	(0.055)
γ	0.791	ζ	0.733				
γ s.e.	(0.046)	ζ s.e.	(0.062)				
J	45.884						
p-val	0.277						

This table shows the results of the GMM estimation of α and the parameters ρ that make up τ. The estimation is carried out using the continuously updated GMM estimator of Hansen et al. (1996). Column (1) shows the estimates of the production function coefficients α while the remaining columns show the estimates of the coefficients ρ parameterizing τ for each input. The bottom of the table also reports the returns to scale parameter γ and money flexibility parameter ζ and their standard errors. The J statistic and p value from the test of the 42 overidentifying restrictions (92 moment conditions, 50 parameters) are also reported below. I fail to reject the null that all the moment restrictions are valid.

4.2.1 TFP

I recover the TFP residual from the production function estimates:

$$\psi_{it+1} \equiv y_{it+1} - \sum_k \hat{\alpha}_k q_{ikt} \tag{13}$$

and decompose it into a household fixed effect $\bar{A}_i$ plus deviations. Many papers in the misallocation literature assume that productivity is time-invariant within households—that is, all of ψ_{it+1} beyond the fixed effect is unanticipated—and stop here. This is a strong assumption, especially in a 16-year panel where households may experience persistent changes in soil quality, management ability, or local conditions.

I relax this assumption by projecting ψ_{it+1} onto variables in households' time-t information esets in order to distinguish the anticipated component a_{it} and the unanticipated component ϕ_{it+1}. If a component of TFP were truly unanticipated, it should be uncorrelated with variables known at planting. Conversely, any predictable component should show up in consumption and investment behavior—under incomplete markets, households respond to anticipated shocks by adjusting savings, borrowing, and input purchases.[25] I attribute the predict component to anticipated TFP ($\hat{a}_{it} = \exp(\bar{a}_i + \hat{\xi}_{it})$) and the residual to the unanticipated shock $\hat{\phi}_{it+1}$. This ensures that $\hat{\phi}_{it+1}$ is mean-zero from the household's ex-ante perspective. This is similar in spirit to the use of $g(z_{it-1})$ to proxy for anticipated *changes to* productivity in (9), except I am now trying to predict the *level* of productivity at time t—in other words, I am trying to predict $h(a_{it-1}) + \xi_{it}$ rather than $h(a_{it-1} - a_{it-1})$. Moreover, I am interested in the predictions themselves, rather than trying to control for them to estimate coefficients.

4.2.2 Λ

The financial wedge Λ_{it} is household i's subjective expectation, as of time t, of the product $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$. Rational expectations allowed us to identify the τs from (11), by guaranteeing that *on average* the realized value of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ equals Λ_{it}. However, any given observation contains both the ex-ante Λ and the realization of ex-post shocks. Attributing these ex-post

[25] For example, if a household anticipated a positive TFP shock, it should adjust consumption (in a direction determined by the intertemporal elasticity of substitution) and, if possible, borrow to finance additional inputs. Under binding borrowing constraints, we might also expect substitution away from other investments. Note that this does not conflict with the identifying assumption for the production function moments because different periods' information sets are being considered. The production function assumes that time t shocks are unpredictable using the prior *year*'s information (implicit in the AR(1) assumption, which I fail to reject), while this procedure allows for some of *this year's* TFP shock to be known prior to planting (for example if the farmer receives a rainfall forecast or falls ill prior to planting.)

shocks would severely inflate estimates of misallocation (Gollin & Udry, 2021). Therefore, I need to isolate the predictable component of the realized value of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ in order to obtain a valid measure of financial distortions.

To do so, I once again project the realized value at $t+1$ onto the household's time t information set. Given that $\mathbb{E}_t[\varphi_{it+1}] = 1$ by construction, any variation in $\hat{\Lambda}_{it}$ reflects either a predictable wedge between marginal utilities at planting and harvest (credit market imperfections) or the dependence of λ_{it+1} on φ_{it+1} (uninsured risk).

The validity of this approach rests on two assumptions. First, is the maintained assumption of rational expectations: households' subjective expectations equal the true conditional expectation given available information. Second, the econometrician observes a sufficiently rich subset of the variables in households' information sets. The Townsend Thai data is exceptionally suited for this—I observe monthly consumption, asset holdings, income from multiple sources, credit access, and other variables that plausibly enter households' forecasts.

4.2.3 Implementation

For both projections, I use random forests rather than linear regression. This choice reflects the fact that households may use information in nonlinear ways that are difficult to specify ex ante—for example, the relationship between lagged consumption growth and expected productivity may differ at different wealth levels. Random forests accommodate such nonlinearities while guarding against overfitting through cross-validation.

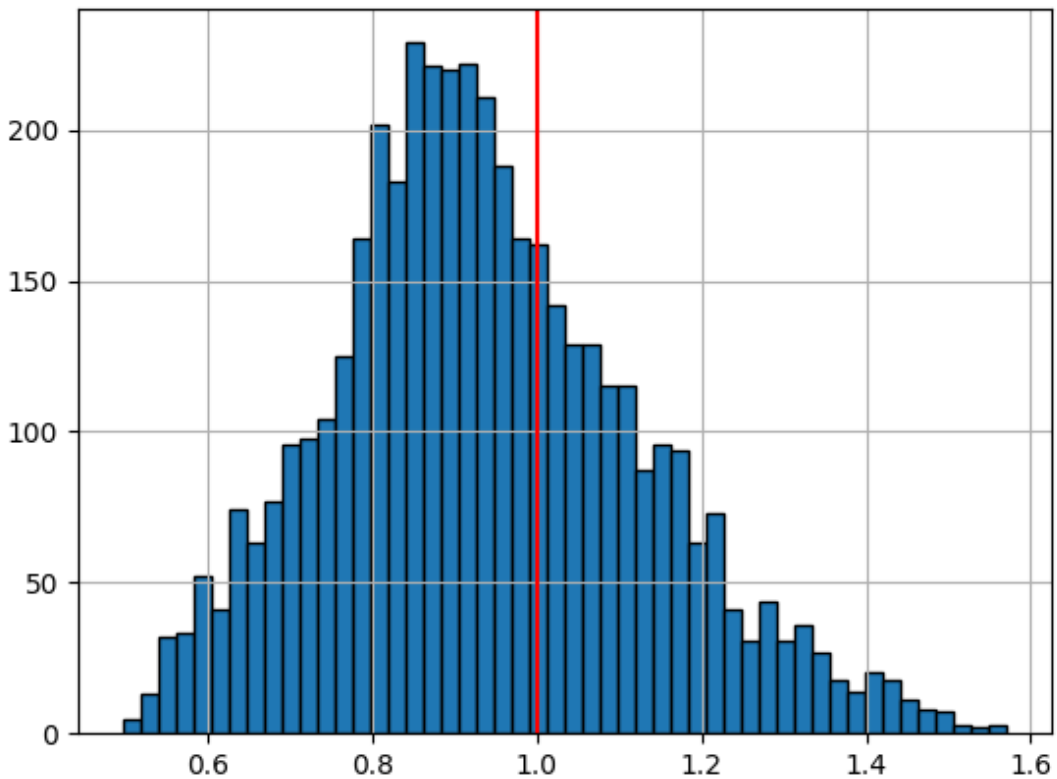
An important caveat is that I can only project onto *observed* variables in the information set. If households have private information that predicts productivity or marginal utility but is not captured in the survey, the "unanticipated" component $\hat{\varphi}$ will contain some forecastable variation, and $\hat{\Lambda}$ will be measured with error. This would attenuate the estimated contribution of financial frictions to misallocation, making my estimates conservative. [26]

After purging ψ_{it+1} of it's time-average $\bar{a}_i$, the random forest is only able to predict about 3.3% of the remaining variation in ex-post realized TFP. $\hat{\xi}_i$ has a mean of -0.007 and a standard deviation of 0.06. In contrast, $\hat{\phi}$ is mean 0.007 and has a standard deviation of 0.38 (0.28 after winsorizing the 1% tails). So household only appear able to forecast a small component of productivity shocks, at least as reflected in their ex ante consumption and investment.

The random forest predicts the realized $\frac{\lambda_{it+1}}{\lambda_{it}}$ with an R^2 of about 0.2, suggesting that households anticipate that future production shocks will cause a great deal of variance in

[26] Note that even if I perfectly observed households' information sets and knew their beliefs, I would not obtain an R^2 of 1 for any of these predictions, as some of the variation in the target variable is fundamentally unpredictable from the *household's* perspective, not just the econometrician's.

Figure 1: Histogram of Estimated Financial Distortions $\hat{\Lambda}$

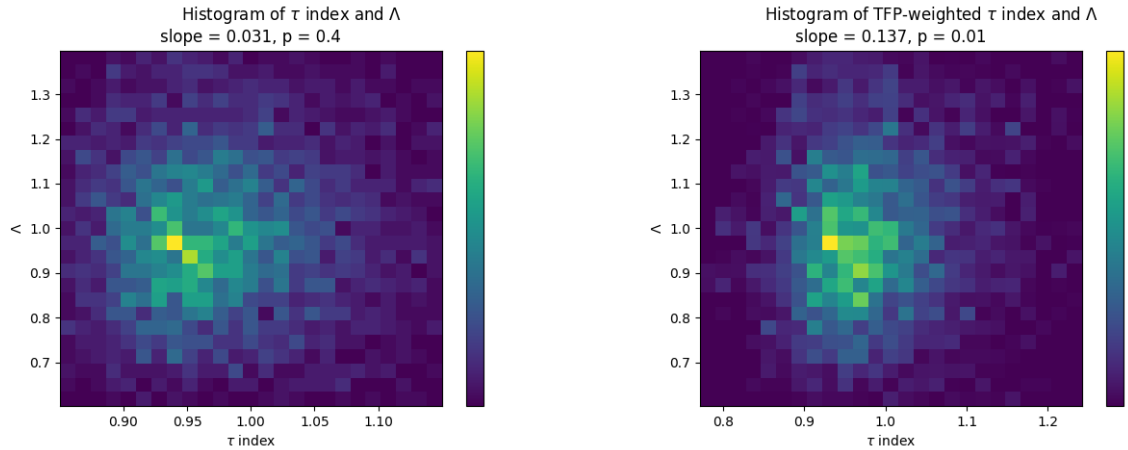


The figure shows the distribution of the esitmated financial distortions $\hat{\Lambda}$ from the main random forest specification.

consumption growth. Naturally, this distribution of the *predicted* $\frac{\lambda_{it+1}}{\lambda_{it}}$ is a shrunken version of the distribution of the *realized* $\frac{\lambda_{it+1}}{\lambda_{it}}$, as seen in Figure C6. Figure 1 shows the distribution of estimated financial wedges $\hat{\Lambda}$. The mean (median) of $\hat{\Lambda}$ is 0.936 (0.919) and the distribution is fairly tight, with a standard deviation of 0.192 and an interquartile range of 0.807–1.058. While this is close to the perfect markets benchmark of 1, (6) shows that $Λ$$ enters input demands raised to the elasticity $\eta \approx 4.79$, meaning that the median household is only producing at about two-thirds of its desired scale due to credit and risk, given prices and input wedges.

Note that about two-thirds of households have a $\hat{\Lambda} > 1$, which suggests that agriculture is a hedge against the rest of the household's portfolio. This is indeed most prevalent in the wealthier and more urbanized province Chachoengsao, where agriculture makes up a smaller share of household's income.

Figure 2: Joint distribution of τ and Λ



This figure plots TFP-weighted histograms of Λ and τ in 25×25 bins and reports the slope coefficients. The left panel plots the Cobb-Douglas price index of τs, $\prod_l \tau_{ilt}^{\alpha_l}$ against the estimates of financial distortions Λ, and the right panel plots the correlation weighted by estimated TFP A, which corresponds to the model-consistent moment (see 6).

5 Decomposition of Misallocation

While the results of the previous section showed the *marginal* distributions of $\hat{\Lambda}$, $\hat{\tau}$ and $\hat{A}$, what matters for aggregate productivity and misallocation, according to Section 3is their *joint* distribution.

[27] Before delving into the decomposition of misallocation, I provide some descriptive graphical evidence to characterize this distribution.

5.1 Descriptive Results

Figure 2 plots 2D histograms of input and financial distortions and reports their correlation coefficients. The left panel plots the unweighted Cobb-Douglas price index of τs, $\prod_l \tau_{ilt}^{\alpha_l}$ against the estimates of financial distortions Λ, while the right panel plots each weighted by TFP A. The positive correlation between τ and Λ suggests that, on average, more financially constrained households are relatively *subsidized* on inputs, although notably this is much larger and statistically significant when weighting by TFP. As illustrated by 6, the product of different distortions is weighted by TFP in equilibrium, reflecting that more productive households command a larger share of inputs. This corresponds to the conventional wisdom that poorer households oversupply labor to their own farms under imperfect labor markets (Breza et al., 2021; Jones et al., 2022; LaFave & Thomas, 2016).

[27] This is an extension of results in Hsieh and Klenow (2009) and Adamopoulos et al. (2022) regarding the covariance between wedges as a sufficient statistic for misallocation.

This implies that the observed distortions partially offset each other — relaxing credit constraints would disproportionately direct capital toward farms that are effectively subsidized on inputs. The direct gains from relaxing credit constraints are large enough to swamp this effect but are smaller than they would be if credit constraints were uncorrelated with input distortions.[28] The results also show that distortions for land and labor are positively correlated. Most of the misallocation literature rules this out by assumption, modeling τ as a distortion in the *relative* price of land and labor. However, I am able to relax this assumption by using fertilizer and seed as normalizing inputs when estimating τs.

5.2 Main Decomposition

I now decompose the productivity dispersion in the Thai data into four parts: misallocation due to imperfect financial markets, misallocation due to imperfect factor markets, their interaction effects, and a residual that cannot be interpreted as misallocation according to the model. I do this by computing *expected* aggregate productivity under the following four scenarios: (1) the first-best allocation; (2) the baseline allocation, with all of the distortions I measure; (3) an allocation with perfect financial markets and the observed input wedges; (4) an allocation with perfect input markets and the observed financial wedge. I compute aggregate TFP under these allocations within township-years and then sum up these gains across townships in each of the 16 years of the sample.

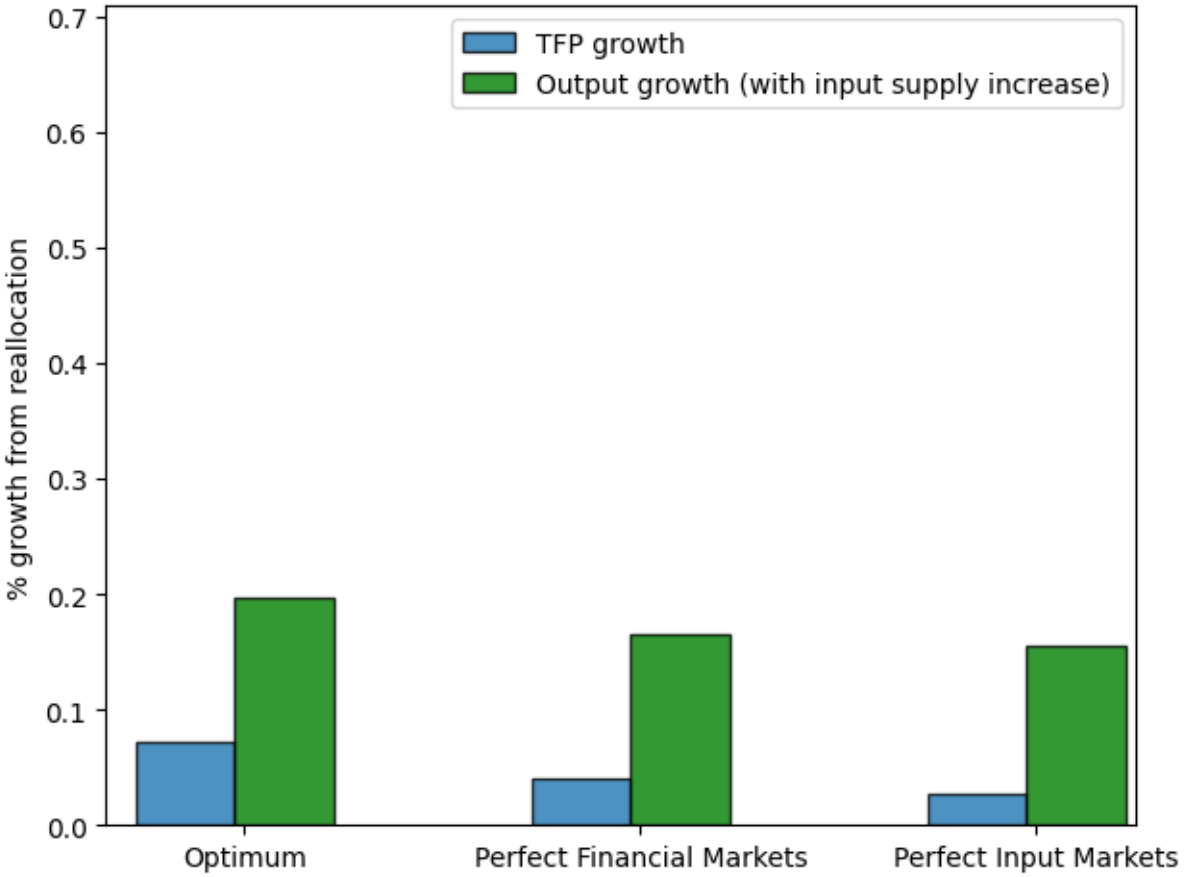
I provide four main sets of results. First I characterize overall misallocation in Thailand. Second, I decompose misallocation into input distortions, financial distortions, and interactions between them. I then show other methods that are more susceptible to measurement error in inputs yield starkly different results. Finally, I use the model to approximate the marginal returns to incremental reductions in one or both sets of distortions. Note that the results below all refer to expected TFP, since the realizations of ex-post shocks cannot be considered misallocation.

The gains from reallocation depend on whether one assumes that the stock of tradable inputs is held fixed or can respond to changes in demand. The results also depend on whether one assumes input frictions take the form of implicit taxes or rations. I show how results depend on each of these cases below.

Baseline Misallocation Figure 3 plots the gains from reallocation under three scenarios as a percentage of (expected) aggregate TFP in the observed allocation: (1) eliminating

[28] TFP governs the incidence of these distortions; since it is the sole determinant of scale under the efficient allocation, multiplicative wedges such as Λ or τ exert a large influence on the aggregate economy when it affects firms that command more inputs.

Figure 3: Decomposition of TFP gains from reallocation



The figure shows the decomposition of misallocation summed up across years, as a percentage of status quo aggregate TFP. The first group of columns shows results under perfect financial markets but with the observed input frictions. The second shows results under perfect input markets but with the observed financial distortions. The third shows the results under a full set of perfect markets. The blue (left) bars in each group show the gains holding aggregate supply fixed at the township level for all inputs while the green (right) bars show the gains allowing the aggregate supply of seed, fertilizer, and equipment to increase (holding their prices constant). The results are computed, CFE demands, and restricting the sample to rice plots aggregated to the farm level.

financial distortions (i.e. setting $\Lambda = 1$) holding input frictions τ fixed; (2) eliminating input distortions (setting $\tau = 1$) while holding Λ fixed; and (3) eliminating all distortions. The difference between (1)+(2) and (3) corresponds to the interaction between input and financial misallocation, which can be positive or negative. The blue (left) bars show results holding the aggregate supply of all inputs fixed, as if villages are in autarky. In this case, aggregate TFP is directly proportional to aggregate output. This is a relatively conservative assumption because it excludes gains from the increased aggregate demand for tradable inputs. The green (right) bars allow intermediate inputs (fertilizer, seed, and equipment) to be imported from outside the village at a constant price (as if the village were a small open economy). Confidence intervals from 200 bootstrap replications are shown for each specification.

The gains from full reallocation are 7.2% in the baseline case and 19.7% when the aggregate supply of tradable inputs is allowed to adjust. The baseline estimates are similar to Shenoy (2017)'s estimates from Thailand, which I discuss below. On the other hand, my results are an order of magnitude lower than some estimates from Africa of up to 286% gains from reallocation Aragon et al. (2022) and Chen et al. (2023). The additional gains from allowing the aggregate supply of tradable inputs to adjust are much smaller than those in Carrillo et al. (2023), where they account for almost all the estimated misallocation.[29]

It is clear from the first two groups of bars in Figure 3 that both sets of markets contribute significantly to misallocation in isolation. Perfecting financial markets while holding observed input distortions intact achieves about 55.3% of the possible efficiency gains, or 4.0% of observed TFP. Similarly, removing input distortions holding observed financial frictions intact achieves about 38.6% of these gains 2.8% of TFP).

Notably, these two gains sum to less than 100%, meaning the gains from full reallocation are more than the sum of its parts. This is because Λ and τ are positively correlated (when weighted by TFP). In other words, the most financially constrained households are relatively subsidized in input markets, especially labor, as shown in Figure 2.[30] The effect of relaxing financial constraints is thus attenuated — but not offset — by reallocating resources to farms made inefficiently large by other distortions.

Overall, these patterns suggest that the effects of policies targeting a single market failure would be attenuated, rather than amplified, by failures in other markets.

I further decompose factor misallocation into that resulting from frictions land, equipment, and total labor as well as labor at different points in the season. Table 2 shows the results of removing wedges from each of these markets, with and without relaxing financial constraints. Reducing frictions in labor markets is slightly more effective than for land mar-

[29]See Donovan (2021) for a more detailed discussion of this channel where the price of intermediates is endogenous in general equilibrium.

[30]This reflects the common finding that poorer households tend to oversupply labor to their own plots.

Table 2: Decomposition of Gains by Input Market

	With Financial Constraints	Perfect Financial Markets
Equip.	0.003	0.044
Fert,	0.000	0.040
Harv. Labor	0.020	0.062
Land	0.002	0.043
Plant. Labor	0.001	0.042
Seed	0.000	0.041
Weed. Labor	0.001	0.041

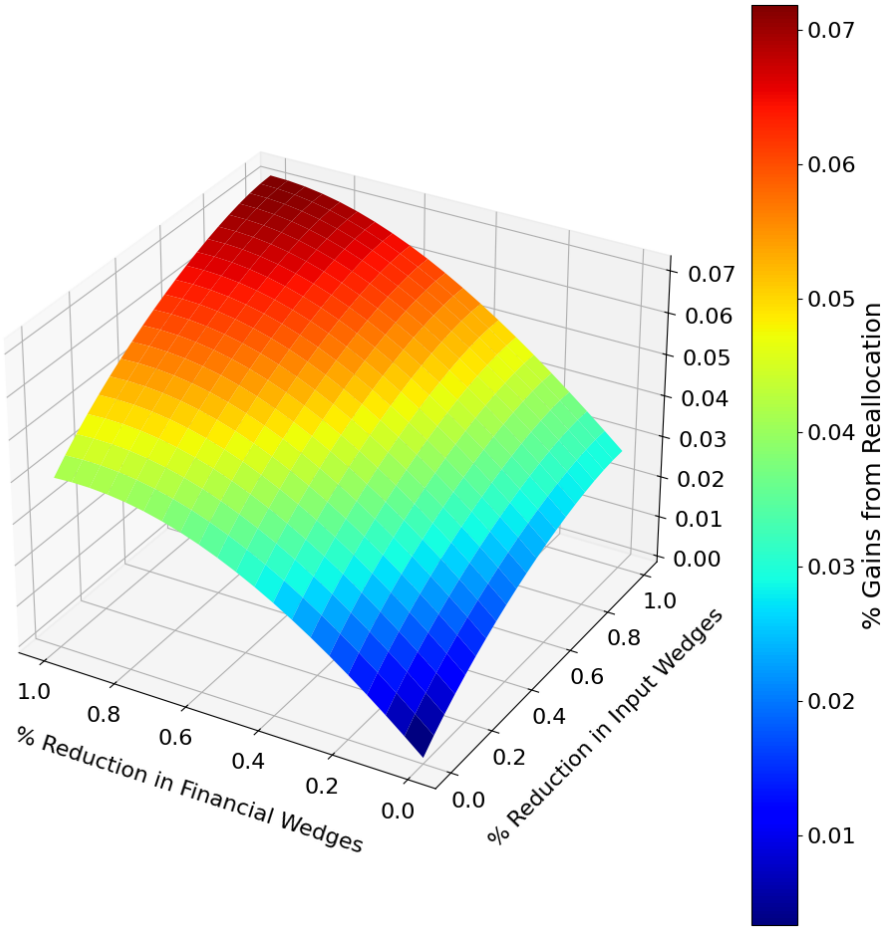
This table shows the gains from removing distortions τ_{ikt} in individual input markets, both with the observed financial constraints and under perfect financial markets. This is shown for the closed economy casem CFE demands, and restricting the sample to rice crops at the farm level.

kets, despite them accounting for roughly equal expenditure shares. The sum of gains from reducing individual frictions is also more than the gains from reducing all of them simultaneously. While input frictions are negatively correlated with financial distortions, they are positively correlated with each other. In other words, reducing frictions in land markets also indirectly addresses labor market distortions by reallocating resources toward households that are relatively taxed.

Intermediate Policies The results above consider the gains from completely eliminating one set of distortions while holding others fixed at observed values. However, policymakers likely have a menu of policy instruments to choose from, but may not be able to fully eliminate distortions. The model allows me to estimate aggregate TFP under any values of Λ and τ. I therefore conduct a simple illustrative exercise in Figure 4, in which I plot the TFP gains from uniform partial reductions in τs and λs. This approximates the marginal returns to reductions in distortions. However, modeling the effects of a specific policy would require assumptions on the specific institutions underlying the distortions I measure, which also govern the second-order effects of how a change in τ affects Λ (and vice versa).

Figure 4 illustrates the complementarities between policies that reduce both sets of distortions. In particular, it shows that the marginal returns to reducing either set of distortions alone are limited, moving along either horizontal axis. However, the marginal returns are much higher after both sets of distortions have been reduced substantially, suggesting that small reductions to one or both sets of distortions may have limited effects and that significant improvements to both sets of markets may be required to unlock large gains. If one knew the relative costs of reducing each distortion, the gradient of Figure 4 would define an expansion path for the social planner in terms of which distortions to target as its budget

Figure 4: Gains from partial reductions of τ and Λ



The figure shows the decomposition of misallocation using the TFP-based measure under different reductions of input and financial wedges. I compute aggregate TFP under each scenario shrinking Λ and τ towards unity by increments of .05. The origin corresponds to the status quo allocation and (1,1) corresponds to the efficient allocation. The vertical axis shows the percent increase in aggregate TFP relative to the status quo allocation. The figure uses CFE demands and restricts the sample to rice plots, aggregating to the farm level.

shifts out. Additionally, Figure 4 shows that these marginal returns are not monotonic: at baseline levels of input (financial) distortions, going from 10% of observed financial (input) distortions to perfect financial (input) markets actually worsens efficiency.

5.3 Methodological Differences and Measurement Error

I now describe how estimating both Λ and τ helps alleviate concerns about measurement error. With both Λ and τ, aggregate productivity under a given allocation can be computed in two ways: taking the observed allocation and then "removing" a distortion or taking the first-best allocation and "adding a distortion". To see this, note that the efficient allocation

(7), which is just a function of A_{it}, can also be written as a function of observed input demands and wedges by inverting (5) as a function of A and dividing out constants

$$\omega_{it}^{*} = \frac{q_{ikt}\tau_{ikt}\left(\frac{\prod_l \tau_{ilt}^{\alpha_l}}{\Lambda_{it}}\right)^{1-\gamma}}{\sum_{h=1}^{N_{vt}} q_{ikt}\tau_{ikt}\left(\frac{\prod_l \tau_{ilt}^{\alpha_l}}{\Lambda_{it}}\right)^{1-\gamma}}. \tag{14}$$

Likewise, under the status quo, rewriting (6) should simply yield

$$\omega_{ikt} = \frac{q_{ikt}}{\sum_{h=1}^{N_{vt}} q_{ikt}} \tag{15}$$

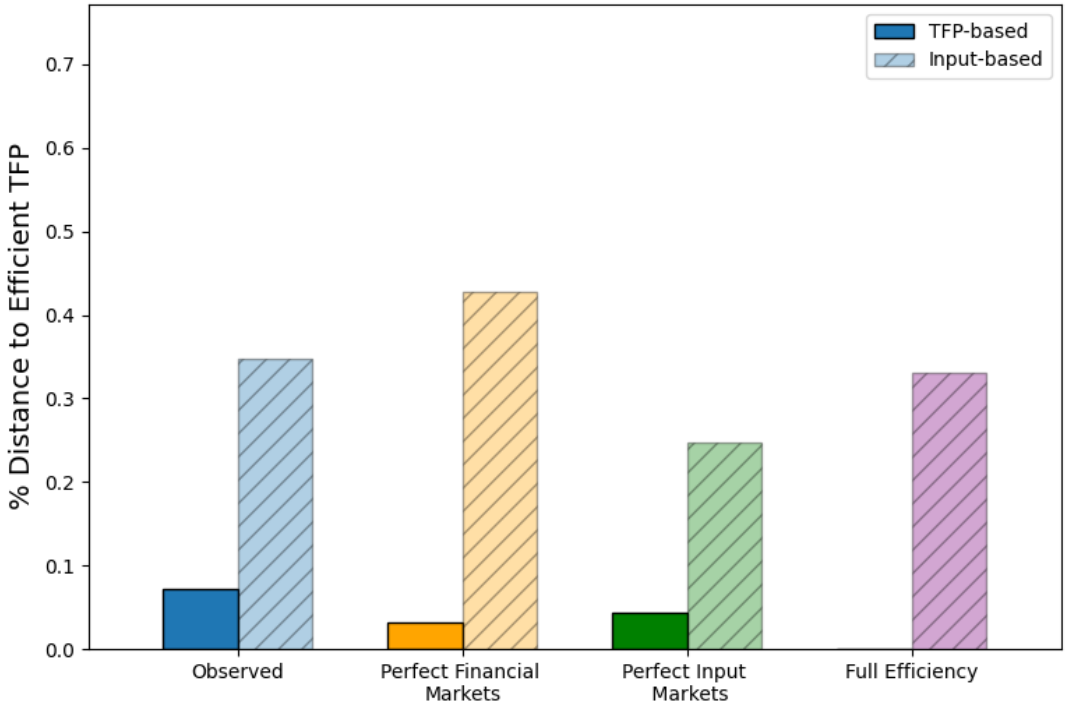
This allows me to compute TFP using either (6) or (14) and then aggregate for any values of Λ and τ. However, this requires estimates of both Λ and τ.

If inputs were measured perfectly and τ and Λ were estimated without error, then these two approaches should produce identical estimates. The difference is that the former approach (6 and 7) uses estimated TFP while the latter (14 and 15) uses raw input measurements. Which estimate is preferable depends on how severe measurement error in inputs is relative to the errors in estimated objects. Given that estimates of TFP are less noisy than the raw inputs used to estimate them, one would therefore expect estimates using the TFP-based measures in (6) and (7) to be more reliable than the input-based measures in (14) and (15).

How different are the conclusions these measures produce in the data? To make this comparison, it will be useful to denominate misallocation by the *attainable* output (equivalent to TFP when aggregate input supply is fixed) forgone due to distortions in each scenario. Figure 5 compares results from the TFP-based results in the solid bars and the input-based results in the shaded bars. The solid bars simply recast the estimates from Figure 3. The blue bars show the percent of attainable output foregone in the observed allocation, while the orange (green) bars show allocations with only the observed input (financial) frictions. By definition, the optimum allocation achieves all the attainable output so there is no solid purple bar.

Now contrast these TFP-based results with the shaded bars, which are computed using the input-based measure. As discussed in Section 3, these two panels would yield identical results if there were no measurement error and the model was perfectly specified. However, the differences between the two panels are quite striking when comparing bars of the same color in Figure 5. First, measured misallocation in the status quo is similar when using the input-based rather than the TFP-based measure, albeit much noisier. Second, it appears that perfecting financial markets would nearby *double* misallocation, rather than decreasing it.

Figure 5: Aggregate TFP relative to optimum, model- vs. input-based measures



The figure shows the percentage of foregone attainable output from the four main allocations considered (the observed allocation, efficient allocation, perfect financial markets with input wedges intact, and perfect input markets with financial wedges intact). The solid bars compute these using the TFP-based measure of misallocation, using (6). The shaded bars are calculated by taking raw input observed in the data and augmenting them by the estimated τ and Λ, where relevant. 95% confidence intervals from 200 bootstrap replications are plotted. Results are computed using CFE demands, only rice plots, and aggregating to the farm level.

Most strikingly though, the implied "optimum" allocation is not only suboptimal but actually performs worse than the observed allocation.

How is this possible? Recall that decompositions using the input-based measure are computed by adding distortions to the observed allocation, which includes mismeasured inputs. The shaded green bar is calculated by equalizing factor ratios in a way that preserves scale across farmers: this is the model of an exchange economy that serves as a lower bound on factor misallocation in Shenoy (2017). The purple bar is then calculated by reweighting those demands by $1/\Lambda$, removing estimated financial frictions.[31] The input-based estimates are higher across the board than those using only estimated quantities. The conflicting result that removing financial frictions would worsen misallocation can be explained by their negative correlation with input measurement error. In other words, measurement error looks like a distortion that is partially offset by financial frictions — removing financial wedges thus makes this spurious distortion appear worse.[32]

These differences between the TFP and input-based measures are quite robust across specifications. Together, these results underscore the importance of separately identifying both input and financial distortions. Without a credible estimate of financial distortions, one would need to rely on noisily measured inputs and arrive at qualitatively different conclusions about the effects of market interventions.

5.4 Alternative specifications and robustness checks

While the magnitudes of misallocation differ slightly across specifications, the qualitative results are broadly consistent.

5.4.1 Taxes vs. Rations

While the estimation procedure doesn't require taking a stand on whether input wedges operate as taxes or rations, this affects how households would adjust input demands in responds to changes in the wedges. In particular, a household facing a downward labor ration, as in Breza et al. (2021), would not use additional credit to hire more labor. The results in Figure 3 treat all inputs as flexible, as if input frictions functioned as taxes.Figure C2 shows the decomposition of gains from reallocation if land were a fixed factor or labor were rationed

[31] Note that the same wedges are used in each set of results but for different specifications. Input wedges are used to compute the orange and blue solid bars and the green and purple bars in the right panel. Meanwhile, financial wedges are used to compute the blue and green solid bars and the purple and orange shaded bars.

[32] Arthi et al. (2018) find that labor inputs are more upwardly biased for smaller farms. Since Figure 2 shows that these households are more financially constrained, financial constraints would then be negatively correlated with the measurement bias. Relaxing these constraints would therefore allocate more resources to farms that appear artificially large in the raw data.

from below, relative to the case where both factors are mobile yet subject to distortions. The blue (left) bars in each group reproduce the results from the baseline case of Figure 3. The green (middle) bars show the results assuming land is a fully fixed factor and the purple (right) bars show results assuming all households with $\tau < 1$ for labor face a downward ration. The inability of land to adjust reduces the potential gains from reallocation by almost half when removing all other market failures, more so when removing only financial frictions and less so when relaxing other input frictions. However maintaining the labor ration has much more muted negative effects and actually leads to even larger gains when keeping financial frictions intact — again owing to the fact that labor and financial frictions are negatively correlated. Even though households facing a downward labor ration would use additional credit to acquire other inputs until the ration no longer binds, the price of these other inputs also increases in equilibrium.

5.4.2 Levels of aggregation

So far I have assumed that reallocation occurs within townships, in which stocks of land and labor are fixed. I argue that this is a realistic level of aggregation since village boundaries within townships are fairly arbitrary Kaboski and Townsend (2011). However, I now consider how these results would change if reallocation could only occur within villages, or if reallocation could also take place across regions of Thailand. The latter should be viewed as an upper bound on the gains from reallocation since fundamental trade and migration costs cannot be considered misallocation. However, if these gains are large, it suggests that investments in roads and other infrastructure that promotes market integration may be effective at reducing misallocation.

Figure C5 shows the potential gains from full reallocation if allocation only occurs within villages or occurs at the national level.[33] The gains from reallocation across regions are more than three times as large as those from reallocation within townships. However, there appears to be very little misallocation across villages within townships, consistent with other evidence that villages in the same area are fairly integrated.

6 Conclusion

In this paper, I estimate distinct distortions affecting farm households in Thailand and quantify how they each contribute to misallocation. This is necessary for policymakers to consider, as the welfare effects of interventions in a single market are ex-ante ambiguous. First, the

[33]Note that since only 16 villages from 4 tambons are included in the sample, this should not be considered representative of a national-level reallocation.

model yields a novel, theory-consistent production function estimation approach that holds when input choices are distorted. My approach flexibly allows for TFP shocks unobserved to the econometrician. Empirically, I find relatively low levels of misallocation in Thai agriculture: In my preferred specification, the estimated gains from optimal reallocation are 7.2%. I further decompose these gains into 2.8% from correcting financial misallocation alone, 4.0% from correcting factor misallocation alone, and the remaining 0.4% from the compounding effects of these distortions. Factor and financial misallocation compound each other because more financially constrained farmers appear relatively subsidized in input markets, particularly for labor. This suggests that policies that seek to alleviate both distortions may be more effective than those targeted towards a single one.

Directly estimating financial distortions rather than inferring them from a residual allows me to avoid attributing measurement error in inputs to misallocation. I find that not accounting for measurement error using the full model would lead to over four times larger estimates of misallocation and, in contrast to my preferred approach, suggest that removing financial frictions alone would worsen misallocation. While the model explicitly allows for such a possibility, my preferred results show that this is not the case.

This paper leaves many additional topics for future research. In particular, more work is required to understand the distributional implications of productivity-enhancing policies. Another open question is how misallocation in agriculture interacts with climate change, given that it increases production uncertainty but increasing agricultural production may create climate externalities. Finally, while the paper provides a broad framework for diagnosing the effects of a general set of distortions, more research is needed to understand specific policies to address the relevant institutions in different contexts.

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A Full Dynamic Program

A.1 Production

Production is given by

$$Y_{it+1} = F(q_{it}, \varphi_{it+1}) \tag{16}$$

where q_{it} is a vector of K inputs applied by j at time t, and φ_{it+1} is a shock realized at $t+1$, prior to harvesting output Y_{it+1}. As is standard, I assume that $F_k > 0, F_\varphi > 0$,and $F_{kk} < 0$ for each k. I assume that F is common across households and fixed over time, but households may have heterogeneous time-varying productivity. Note that I treat all inputs as static – in a benchmark economy with complete rental markets, households' input use at time t would not depend on their endowments or previous seasons' input choices.

I assume that $\bar{w}_{vkt}$ is the (endogenously determined) market price of each input k in village v at time t. However, households may face idiosyncratic taxes or subsidies such that they face prices $s_{ikt}\bar{w}_{vkt}$. Households may also be subject to upward or downward rations on inputs such that $\underline{q}_{ikt} \leq q_{ikt} \leq \bar{q}_{ikt}$.

While I only directly model the agricultural sector, allowing households to earn income from other sources is important to match the income diversification observed in the data. Households can invest in a portfolio of assets b_{imt} with uncertain returns r_{imt+1}. They may also be subject to borrowing constraints such that $\sum_m b_{imt} \geq \bar{B}_{it}$. B should also be thought of as capturing formal and informal insurance with state-contingent payouts. Like with inputs, frictions in the asset market can be modeled by writing returns as $r_{imt+1} \equiv \chi_{imt}\bar{r}_{vmt+1}$, where $\bar{r}_{vmt+1}$ is the (endogenously determined and possibly stochastic) average return in village v.[34] Let B_{it} denote a household's portfolio of assets and R_{it+1} be the return to that portfolio. I denote the set of primitive taxes and rations that generate the distortions I derive below as $\mathcal{D} \equiv \{\chi, s, \underline{q}, \bar{q}, \bar{B}\}$.[35] Note that the estimation strategy I develop in section 4 does not depend on which frictions in $\mathcal{D}$ generate λ and τ. In section 5, I discuss how whether input frictions act as taxes or rations affects the decomposition and compute results both ways.

A.2 Dynamic Program

I assume households j have time-separable, von Neumann-Morgenstern preferences with discount factor δ and per-period utility function $u(c, l)$, which I assume is continuously differentiable, strictly increasing, and concave in consumption c and leisure l. At time t, they

[34] $\chi_{imt} = -\infty$ implies a household never purchases asset m.

[35] While the elements of $\mathcal{D}$ cannot be separately identified without many additional assumptions, they microfound the distortions the markets in credit, insurance and the k input markets I derive below.

maximize

$$\mathbb{E}_t \left[\sum_{s=t}^{\infty} \delta^{s-t} u(c_{is}, l_{is}) \right]$$

subject to the following budget constraint,

$$M_{it+1} = M_{it} + Y_{it+1} - w'_{it} q_{it} - p'_t c_{it} + R_{it+1} B_{it+1} - B_{it} \tag{17}$$

which holds in each state of the world.

The household's value function satisfies the Bellman equation

$$V(Y, M, w, p, \varphi, R, \mathcal{D}) = \max_{c,q,B} u(c) + \delta \mathbb{E}_t V(Y', M', w', p', \varphi', R', \mathcal{D}') \tag{18}$$

subject to the budget constraint (17), borrowing constraint $\bar{B}$, and possible rations on hiring inputs in or out, $\underline{q}, \bar{q}$. Taking first-order conditions with respect to the choice variables c, q, and B:

$$(c) \qquad u_i(c) = \lambda p_i \tag{19}$$

$$(q) \qquad \delta \mathbb{E} \left[\frac{\partial V}{\partial Y} (Y', k', w', p', \varphi', R', \mathcal{D}) F_k(q, \varphi') \right] = \lambda w_k + \underline{\mu}_k - \bar{\mu}_k \tag{20}$$

$$(B) \qquad \delta R \mathbb{E} \left[\frac{\partial V}{\partial B} (Y', k', w', p', \varphi', R', \mathcal{D}') \right] + \mu^B = \lambda \tag{21}$$

where λ, μ^B, $\underline{\mu}_k$, and $\bar{\mu}_k$ are the Lagrange multipliers on the budget constraint, borrowing constraint $\bar{B}$, and rations on hiring inputs in and out, $\underline{q}, \bar{q}$, respectively. The first FOC simply states that households equate the marginal utility of expenditure on each good consumed within a period to a common Lagrange multiplier λ. The second implies that households equate the marginal utility of expenditure on each input to the expected marginal utility of its marginal product, unless an input ration binds. The third is simply the Euler equation with the possibility of binding borrowing constraints.

A.3 Input Demands and Wedges

Applying the envelope theorem to the first-order condition for q with simple substitutions yields the following expression for input demands:

$$\bar{w}_{vkt} \tau_{ikt} = \delta \mathbb{E}_t \left[F_k(q_{it}, \varphi_{it+1}) \right] \Lambda_{ikt} \tag{22}$$

in which

$$\tau_{ikt} \equiv s_{ikt} + \frac{\underline{\mu}_{ikt} - \bar{\mu}_{ikt}}{\lambda_{it}\bar{w}_{vkt}} \tag{23}$$

$$\Lambda_{ikt} \equiv \frac{\mathbb{E}_t[\lambda_{it+1}]}{\lambda_{it}} + \frac{cov_t(\lambda_{it+1}, F_k(q_{ik}, \varphi_{it+1}))}{\lambda_{it}} \tag{24}$$

(22) simply states that households equate the marginal utility of expenditure on input k to the discounted expected marginal utility of its marginal product. Under input frictions, the (shadow) cost of each input k differs from the common market price by τ_{ikt} as defined by (23). Meanwhile, Λ captures how credit constraints and uninsured risk affect input demands through the two terms in (24), respectively. When credit constraints bind, (Equation 20) implies that $\lambda_{it} > \mathbb{E}_t[\lambda_{it+1}]$ since households cannot borrow against expected future earnings. Likewise, absent full insurance, consumption at $t+1$ will depend on the realization of production shocks, creating a non-zero covariance between λ_{it+1} and (stochastic) marginal products, $F_k(q_{it}, \varphi_{it+1})$. This covariance may differ across inputs for a general production function. However, it will be negative if households are prudent ($u'''(c) > 0$), input k does not reduce risk ($F_{k\varphi} \geq 0$), and agriculture is not a hedge against overall portfolio risk. In this case, both mechanisms would reduce input demands relative to the case of perfect financial markets.

Λ_{ikt} and τ_{ikt} fully characterize the distortions generated by $\mathcal{D}$ in the markets for each input k. To see this, compare (22) to the benchmark of perfect markets, in which it reduces to expected profit maximization.

$$\bar{w}_{vkt} = \delta\mathbb{E}_t[F_k(q_{it}, \varphi_{it+1})] \tag{25}$$

This is identical to (22) when $\Lambda_{ikt} = \tau_{ikt} = 1$ for all j, k, t. In this case, ratios of marginal utilities λ are constant across households and cancel out and all households equalize expected marginal products to the common price of each input ($\tau = 1$). The equalization of marginal products across households implies the allocation is efficient. Note how deviations from efficiency are completely characterized by Λ_{ikt} and τ_{ikt}, which together define the distortions in the market for each input k.

I have thus far kept the model as general as possible to illustrate how financial and input frictions create distinct wedges under very general conditions. However, estimating the model requires functional form assumptions for F and u. While I discuss functional forms for utility in section 4, I assume output is determined by the following Cobb-Douglas production function:

I assume production takes the following Cobb-Douglas form

$$Y_{it+1} = e^{\psi_{it+1}} \prod_k Q_{ikt}^{\alpha_k}. \tag{26}$$

Without loss of generality, I assume $\psi_{it+1} = a_{it} + \phi_{it+1}$, where a_{it} is anticipated productivity and ϕ_{it+1} is an unanticipated shock that cannot be predicted with time t information. I assume further a_{it} follows a first order Markov process — $a_{it} = h(a_{it-1}) + \xi_{it}$ — in which $h(\cdot)$ is a general function capturing the anticipated evolution of productivity and ξ_{it} is the unpredictable "innovation" to productivity, orthogonal to household's prior information sets by construction. I assume decreasing returns to scale with $\gamma \equiv \sum_k \alpha_k < 1$.[36]

Under the Cobb-Douglas assumption. I can rewrite (22) to obtain the demand function for each input k.

$$q_{ikt} = \frac{\delta \alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \frac{\mathbb{E}_t[\lambda_{it+1} Y_{it+1}]}{\lambda_{it}} \tag{27}$$

(5) can also be expressed as

$$q_{ikt} = \frac{\delta \alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \mathbb{E}_t[Y_{it+1}] \Lambda_{it} \tag{28}$$

where $\Lambda_{it} = \dfrac{\mathbb{E}_t[\lambda_{it+1}\varphi_{it+1}]}{\lambda_{it}}$ is now constant across inputs k.[37]

Meanwhile, distortions in the market for each input k enter through τ_{ikt}. In contrast, financial frictions Λ_{it} distort the *scale* of production while the *composition* of inputs is only distorted by τ. To see this, take the ratio of demands for any two inputs, k and l:

$$\frac{q_{ikt}}{q_{ilt}} = \frac{\alpha_k}{\alpha_l} \frac{\bar{w}_{vlt}}{\bar{w}_{vkt}} \frac{\tau_{ilt}}{\tau_{ikt}} \tag{29}$$

Input ratios are solely a function of technology and relative market prices, which under perfect markets are constant across households in the same village-year. Thus any dispersion

[36] If $\gamma \geq 1$, then the efficient allocation is degenerate with only the most productive producer producing.

[37] To see this, it is useful to write the expectation in the numerator as $\mathbb{E}_t[\lambda_{it+1}] + cov_t(\lambda_{it+1}, \varphi_{it+1})$ (since φ is mean 1 by construction). Also note that (28) can be written in closed-form by substituting the production function for Y_{it+1} and solving the system of equations implied by (5)

$$q_{ikt} = \frac{\alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \left(A_{it}\Lambda_{it} \prod_l \left(\frac{\alpha_l}{\bar{w}_{vlt}\tau_{ilt}} \right)^{\alpha_l} \right)^{\eta}$$

where $\eta \equiv \frac{1}{1-\gamma}$

in input ratios can be attributed to τ.[38] This is a feature of any homothetic production function.[39]

A.3.1 Nonhomothetic Production

While the misallocation literature typically assumes a homothetic production function with Hicks-neutral shocks, this implies that all inputs contribute proportionally to the variance as outputs. Maintaining this assumption not only increases tractability, but allows me to directly compare my results to others in the literature, showing how modeling the consumption side produces drastically different conclusion, holding the model fixed. However, a stark implication of homotheticity is that households facing the same input prices would use the same input mix and financial distortions would only affect the scale of production. To relax this assumption, I assume production takes the following generalized Cobb-Douglas form following Just and Pope (1978) and Just and Pope (1979).

$$Y_{it+1} = A_{it} \prod_{k}^{K} q_{ikt}^{\alpha_k} + \varphi_{it+1} B_t \prod_{k}^{K} q_{ikt}^{\beta_k} \tag{30}$$

where Y_{t+1} is output realized the period following production, q_{kt} is the quantity of input k at time t, A is TFP, and φ_{t+1} is a mean 0 shock realized before harvest and consumption at $t+1$. I assume that expected returns to scale $\gamma \equiv \sum_k \alpha_k < 1$ to ensure the socially optimal allocation is nondegenerate. The main difference between this and the workhorse Cobb-Douglas specification is that the variance of output now depends on input composition. Inputs are differentially risky if $\alpha \not\propto \beta$. In particular, α_k can be thought of as the elasticity of the expectation of output with respect to input k, while β_k is the elasticity of the *standard deviation* of output with respect to input k.[40]

$$q_{ikt} = \frac{\alpha_k \mathbb{E}_t[Y_{it+1}]\mathbb{E}_t[\lambda_{it+1}] + \beta_k cov_t(\lambda_{it+1}, Y_{it+1})}{\lambda_{it} \bar{w}_{kvt} \tau_{ikt}} \tag{31}$$

Note how when $\alpha = \beta$ this reduces to (5).

The only difference is that (31) assigns different coefficients to the expected and stochastic components of $\mathbb{E}_t[\lambda_{jt+1} Y_{jt+1}]$. Inputs with higher β contribute more to the variability of output, causing their demand to be disproportionately affected by imperfect insurance. In contrast, the separability of the shocks in the standard Cobb-Douglas means that the same

[38] Note that $s, \underline{q}, \bar{q}, \bar{B}$, and $\{\chi\}$ are the primitives that determine the distortions τ and Λ.

[39] Note that under CES production, the ratio of τs on the right-hand side of (29) is raised to the elasticity of substitution σ.

[40] I prefer this specification to that recently introduced by Bohr et al. (2023), since this functional form allows for a first order effect of uninsured risk on input demand as shown below. Note that this functional form nests the workhorse Cobb-Douglas specification $Y_{t+1} = A_t e^{\phi_{t+1}} \prod_{k=1}^{K} q_{kt}^{\alpha_k}$ if $\alpha = \beta$ and $B = A/\mathbb{E}[e^{\phi}]$.

Λ_{it} applies to demand for each input.[41]

The ratio of λs in the first term can be thought of as the wedge created by the inability to intertemporally smooth consumption and is constant across inputs. For example, if a household faces a binding borrowing constraint, then $\mathbb{E}_t[\lambda_{t+1}]$ would generally be lower than λ_{t+1}. The second term captures how uninsured risk affects demand. Again, one would expect the covariance term to be negative,[42] but this is amplified by how risky a given input is.

However, this no longer allows the straightforward identification of τ from (29), requiring an alternative set of identification assumptions, which I discuss in Appendix B. I also show results from this more general specification and the results are broadly similar to those under the standard Cobb-Douglas. Definition 1 formalizes an equilibrium in either case.

Definition 1. *A decentralized equilibrium is defined by a set of prices* $\{\bar{w}_{vkt}, p_{it}, R_v t\}$*, an input allocation* $\{q_{ikt}\}$*, and a consumption allocation* $\{c_{it}\}$ *such that*

1. *Households optimize following (19)-(21)*
2. *Input demands* q_{ikt} *equal* $\omega_{ikt}\bar{Q}_{vkt}$*, where* ω_{ikt} *is given by (6) and* $\sum_{i=1}^{N_v} \omega_{ikt} = 1$ *for each* v
3. Λ_{it} *and* τ_{ikt} *are defined as in (23) and* (28)

given a set of initial asset holdings M_{it} *and primitive distortions* $\mathcal{D}$*.*

B Generalized Cobb-Douglas Production

B.1 Estimating α and β

I now describe how each of the key components of the model λ, τ, α,β A, and Λ are estimated. Since the estimation of marginal utilities, λ, doesn't depend on the production function, I follow the same procedure as in Section 4.1.2. I then estimate (31) from a subsample of households' input demands. Once I recover α and β, I then recover τ, Λ and A for the full sample of households.

I use GMM to estimate α and β from the system of input demand equations defined by (31). In this approach, inputs are analogous to assets in a CAPM model and λ_t/λ_{t+1} is analogous to the portfolio's return (Hansen & Singleton, 1982). Under rational expectations, this yields a straightforward approach to estimation. a

[41] This is true for any homothetic production function.

[42] unless $u'''(c) \leq 0$ or returns from agriculture are sufficiently negatively correlated with those from other investments

Let $x_{ikt} \equiv \bar{w}_{kvt}\tau_{kt}q_{ikt}$. x_{ikt} can be interpreted as household j's "shadow" expenditure on input k at time t. This can either represent actual expenditure under possibly household-specific prices or as the cost of input k such that the household would choose q_{ikt} under perfect markets. Let $\mathcal{I}_{it}$ be a vector of variables in household j's information set at time t. Rearranging constrained-optimal input demands (31) and making the dependence on households' time t information sets explicit yields the moment condition.

$$\alpha_k \mathbb{E}[\lambda_{i,t+1}|\mathcal{I}_{it}]\mathbb{E}[Y_{i,t+1}|\mathcal{I}_{it}] + \beta_k cov(\lambda_{i,t+1,}Y_{i,t+1}|\mathcal{I}_{it}) - \lambda_{it}x_{ikt} = 0 \tag{32}$$

where $cov_t(\lambda_{i,t+1}Y_{i,t+1}) = \mathbb{E}_t[\lambda_{i,t+1}Y_{i,t+1} - \mathbb{E}_t[\lambda_{i,t+1}]\mathbb{E}_t[Y_{i,t+1}]]$ can be thought of as a measure of how households expect their utility at harvest to depend on the realizations of production shocks, conditional on their time t information. Estimation requires mapping the subjective expectations $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}], \mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$, and $\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$ to data. The nested case of $\alpha = \beta$ in Section 4 doesn't require distinguishing between $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$ and $E[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$, which allows me to substitute realized $\lambda_{jt+1}Y_{jt+1}$ for $\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$ under rational expectations. More formally, Differences between the expected and realized products of output and marginal utilities can be expressed as:

$$\lambda_{i,t+1}Y_{t+1} - \mathbb{E}[\lambda_{i,t+1}Y_{t+1}|\mathcal{I}_{jt}] = \zeta_{i,t+1} \tag{33}$$

I assume households are fully forward-looking and have rational expectations over future shocks. In this case $\mathbb{E}_t[\zeta_{i,t+1}|\mathcal{I}_{jt}] = 0$, as ζ is simply prediction error that arises from the realization of shocks after households' optimal decisions are made in time t. The challenge is that $\mathbb{E}_t[\zeta_{i,t+1}|\mathcal{I}_{jt}]$ is the household's *subjective* expectation as of time t, conditional on its information set $\mathcal{I}_{jt}$ but prior to the realization of shocks, and is unobserved. I do observe $\lambda_{jt+1}Y_{jt+1}$ for N households in T years. In a given year, the *population* the mean of realized ζ_{it+1}, which I denote as $\mathbb{E}[\zeta_{t+1}]$, may be nonzero if there are aggregate shocks that affect all households within a period.

This motivates a set of instruments

$$z_{it} \equiv \{\lambda_{i,t-1}, \lambda_{i,t-2}, \ldots\}$$

that are plausibly orthogonal to prediction error. The logic of this is that past consumption is correlated with future consumption, making z_{it} relevant.[43] However, whether $\mathbb{E}[\zeta_{it+1}z_t] = 0$ depends on whether their covariance is stationary. Within a given year, realizations of

[43]Importantly, these instruments are functions of consumption that takes place after the previous season's shocks are realized, meaning that they don't reflect uncertainty from previous seasons.

shocks are likely to differentially affect households with different levels of wealth, and thus z_t. Intuitively, poorer households may be more risk-averse (under prudence) and less insured, and thus their marginal utilities will be more sensitive to the realizations of shocks. The stationarity of $\mathbb{E}[\zeta_{it+1} z_t] = 0$ ensures that the effects of this greater sensitivity of poorer households averages out to zero in the panel.[44]

This assumption permits the substitution of (subjective) conditional expectations $\mathbb{E}[\lambda_{jt+1} Y_{jt+1} | \mathcal{I}_{jt}]$ with realizations. However, this substitution can't be used for both $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$ and $\mathbb{E}[\lambda_{jt+1} Y_{jt+1} | \mathcal{I}_{jt}]$ Therefore, separately identifying α and β requires taking a stand on what shocks the household does and does not anticipate at time t.

One approach would be projecting realizations of λ_{it+1} and Y_{it+1} on to functions of $\mathcal{I}_{jt}$, say $l(\mathcal{I}_{jt})$ and $y(\mathcal{I}_{jt})$, and using the predicted values, $\hat{l}(\mathcal{I}_{jt})$ and $\hat{y}(\mathcal{I}_{jt})$,to substitute for $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$, respectively. In this case

$$\begin{aligned}
\lambda_{it+1} &= \mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] + \pi^L_{it+1} \\
Y_{it+1} &= \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}] + \pi^Y_{it+1} \\
\lambda_{it+1} &= \hat{l}(\mathcal{I}_{jt}) = +\upsilon^L_{it+1} \\
Y_{it+1} &= \hat{y}(\mathcal{I}_{jt}) + \upsilon^Y_{it+1}
\end{aligned} \tag{34}$$

The household's prediction errors π are mean zero by rational expectations and the estimation errors υ are mean 0 by construction. This means that the difference these two errors $\psi^Y_{it} \equiv \pi^L_{it} - \upsilon^L_{it}$ and $\psi^Y_{it} \equiv \pi^L Yit - \upsilon^Y_{it}$ are each mean zero by linearity of expectations. However substituting the *product* of subjective $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$ for realizations implies:

$$\begin{aligned}
&\mathbb{E}\left[\left(\alpha_k(\hat{l}(\mathcal{I}_{jt}) + \upsilon^L_{it+1})(\hat{y}(I) + \upsilon^Y_{it+1}) + \beta(\lambda_{jt+1} Y_{jt+1} - (\hat{l}(\mathcal{I}_{jt}) + \upsilon^L_{it+1})(\hat{y}(I) + \upsilon^Y_{it+1})) - \lambda_{it} x_{ikt}\right) \otimes h(\mathcal{I}_{jt})\right] \\
&= (\alpha_k - \beta_k)(\psi^L_{it+1} \hat{y}(\mathcal{I}_{jt}) + \psi^Y_{it+1} \hat{l}(\mathcal{I}_{jt}) + \psi^L_{it+1} \psi^Y_{it+1} \otimes h(\mathcal{I}_{jt}) = 0
\end{aligned} \tag{35}$$

Assuming $\hat{l}(\mathcal{I}_{jt})$ and $\hat{y}(\mathcal{I}_{jt})$ provide accurate predictions of the true subjective expectations, $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$, they will differ by estimation error υ_l and υ_y, respectively. Both of these are mean 0 and orthogonal to $\mathcal{I}_{jt}$, by construction. However, their product is not necessarily mean 0. Estimation errors are likely to be correlated absent full insurance

[44]This would be violated if households' mispredictions, conditional on time t information, were correlated with income prior to time t.[45] However, I show with Monte Carlo simulations that the estimator I derive below performs well with small T, even when there is a correlation between baseline wealth and the realizations of aggregate shocks in the finite sample.

or quadratic utility, negatively if $u'''(c) > 0$ and production is not used as a hedge against portfolio risk: any anticipated productivity shock not captured by functions of the observed elements of $\mathcal{I}_{jt}$ is likely to have opposite effects on $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$.

I assume households are fully forward-looking and have rational expectations. Therefore, by virtue of optimization, any differences between realized shocks and households' conditional expectations as of time t are mean-zero prediction errors Denote households' conditional expectations of $\lambda_{i,t+1}$ and $Y_{i,t+1}$, respectively, as $\bar{\lambda}_{i,t+1}$ and $\bar{Y}_{i,t+1}$. This implies that households' subjective expectations of random variables, on average, equal their realizations, meaning that we can substitute the conditional expectations in (32) with their unconditional expectations (Hansen & Singleton, 1982).[46] Letting $\mathbb{E}$ denote the unconditional expectation operator, this implies that estimating

$$\begin{aligned}
&\mathbb{E}\left[\left(\alpha_k \hat{l}(\mathcal{I}_{jt})\hat{y}(I) + \beta(\lambda_{jt+1}Y_{jt+1} - \alpha_k \hat{l}(\mathcal{I}_{jt})\hat{y}(I)) - \lambda_{it}x_{ikt}\right) \otimes h(\mathcal{I}_{jt})\right] \\
&= \mathbb{E}[(\alpha_k(\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}] + \upsilon_{it}) + \beta_k(\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}] - \mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}] - \upsilon_{it} + \zeta_{it+1}) \\
&- \lambda_{it}x_{ikt}) \otimes h(\mathcal{I}_{jt})] \\
&= ((\alpha_k - \beta_k)\upsilon_{it} + \zeta_{xi}) \otimes h(\mathcal{I}_{jt}) = 0
\end{aligned}$$

where $\upsilon_{it} \equiv \upsilon^l_{it}\hat{y}(\mathcal{I}_{jt}) + \upsilon^y_{it}\hat{l}(\mathcal{I}_{jt}) + \upsilon^l_{it}\upsilon^y_{it}$ collects errors from the auxiliary regressions and h is a measurable function of observed elements of $\mathcal{I}_{it}$ and $\otimes$ is the Kronecker product. Intuitively, the difference between households' subjective prediction errors and the econometrician's estimation errors needs to average out to 0 in the sample.

Taking sample averages:

$$g_{NT}(\alpha, \beta) \equiv \frac{1}{NT}\left(\sum_{i=1}^{N}\sum_{t=1}^{T}(\alpha_k - \beta_k)\hat{l}(\mathcal{I}_{jt})\hat{y}(\mathcal{I}_{jt}) + \beta_k(\lambda_{i,t+1}Y_{i,t+1}) - \lambda_{it}x_{ikt}\right) \otimes h(\mathcal{I}_{it}), \quad (36)$$

which converges to 0 with large NT under similiar conditions as in the main analysis. Thus, the GMM estimate of $\theta = \{\alpha, \beta\}$ is

$$\arg\min_{a,b} J(a, b) \equiv g_{NT}(a, b)' W g_{NT}(a, b) \quad (37)$$

where W is the standard optimal weighting matrix.

A second challenge is separately identifying τ, since households facing common technology and prices will no longer necessarily have the same input ratios. To make progress, I draw

[46] Also note that $\mathbb{E}[\lambda_{t+1}Y_{t+1} - \hat{l}(\mathcal{I}_{jt})\hat{y}(\mathcal{I}_{jt})|\mathcal{I}_{it}] = \mathbb{E}[\lambda_{t+1}Y_{t+1} - \hat{l}(\mathcal{I}_{jt})\hat{y}(\mathcal{I}_{jt})]$.

on empirical IO methods to estimate product-level production functions with unobserved input prices. In the case of De Loecker et al. (2016), they observe single- and multi-product firms producing the same goods but only observe inputs at the firm level. Their solution is to estimate the production function restricting the sample to single-product firms, and then apply a selection correction to control for unobservable differences between these two types of firms.

The problem in my case is that τ is not necessarily observed. Depending on the nature of input distortions, τ may correspond to the difference between the market price of an input and the price actually paid by a household that purchases this input, or it may be a shadow price that a household faces when rationed. I observe both input expenditures and quantities in the data. I assume that when households hire labor or rent land, any distortion is reflected in the observed price they pay. In this case, τ_{ikt} is included in the x_{ikt} I observe, which is the appropriate variable for (36). Thus I restrict the sample to transacted inputs when estimating α and β, which I then use to recover τs for the households that do not transact these inputs. Note that I do not have to make such assumptions about the nature of τs when production is homothetic, as I can estimate these directly from factor ratios.[47]

C Additional Tables and Figures

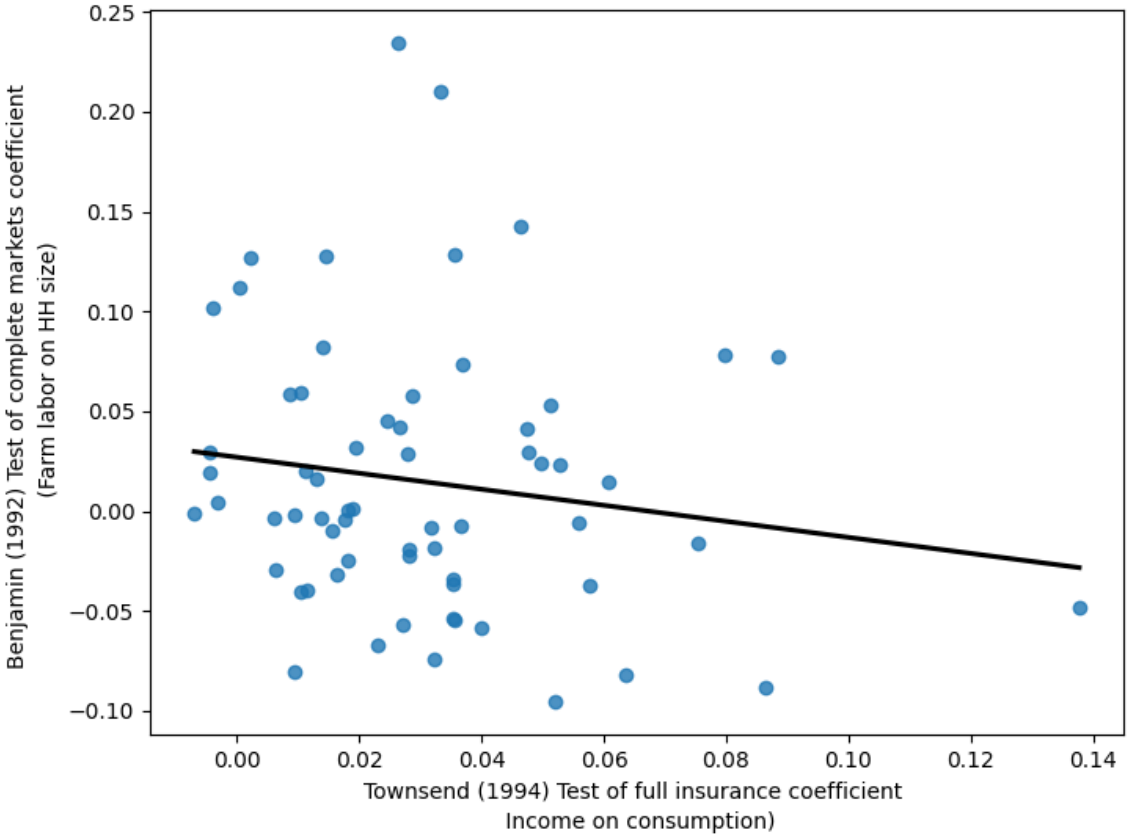
[47]This approach relies on some strong assumptions — namely that there is no selection into hiring inputs, that transacted inputs have the same returns as those owned by the household, and that households who purchase positive amounts of inputs do not come up against a ration. To provide support for the first assumption, I can apply the control function approach in De Loecker et al. (2016). I can also restrict the sample to households that use their own inputs in some seasons and purchase inputs in others. To address the second, I observe individual laborer and plot identifiers and can test whether their observed productivity differs when they are used by their respective households or hired. The third assumption is more difficult to test, but I can attempt to restrict the sample to households that appear less likely to face a binding ration.

Table C1: Diagnostic Tests for Market Failures

	log Consumption Val.	log Labor Hrs.	
	(1)	(2)	(3)
log Income	0.0547***		
	(0.0037)		
HH Size		0.0211*	
		(0.0112)	
Male adults			0.0257
			(0.0258)
Female Adults			0.0269
			(0.0253)
Male children			0.0121
			(0.0217)
Female Children			0.0165
			(0.0210)
Household FE	Yes		
Village-month FE	Yes		
Village-year FE		Yes	Yes
F-stat			11.61**
p-val			0.0205
Observations	83,384	5,689	5,689

This table presents the results for two of the canonical tests of incomplete markets in the literature. Column (1) shows the results of a regression of (log) consumption on income with household and village-month fixed effects as in Townsend (1994). The full monthly sample of households (agricultural and non-agricultural) and monthly measures of total income and consumption are used. Column (2) shows the results of the Benjamin (1992) test of separability, which regresses (log) household labor hours on household characteristics, controlling for farm size. For simplicity, household size is the only measured included. Data from the full sample of producers aggregated to the household-year level are used.

Figure C1: Comparison of Test Coefficients Across Villages



This figure contains a scatter plot of the coefficients from the Townsend (1994) and Benjamin (1992) tests, run separately for each village in each 48-month block of the full panel. For the Townsend coefficients on the x-axis, the full monthly sample of households (agricultural and non-agricultural) and monthly measures of total income and consumption are used. Data from the full sample of producers aggregated to the household-year level are used to estimate the Benjamin coefficients on the y-axis.

Table C2: Summary statistics for agricultural households by township

	All	Chachoengsao	Buriram	Lopburi	Sisaket
HH Size	5.564	5.827	5.622	5.03	5.923
	(2.333)	(2.857)	(2.214)	(2.018)	(2.389)
Age Head	56.037	59.792	53.295	53.756	59.597
	(13.259)	(13.515)	(13.275)	(12.387)	(12.745)
Sex Head	0.804	0.757	0.821	0.842	0.769
	(0.397)	(0.429)	(0.383)	(0.365)	(0.422)
Head Primary Educ	0.87	0.951	0.699	0.948	0.938
	(0.337)	(0.215)	(0.459)	(0.223)	(0.241)
Head Secondary Educ	0.1	0.07	0.08	0.121	0.115
	(0.3)	(0.255)	(0.271)	(0.326)	(0.319)
Formal Loan	0.341	0.149	0.432	0.368	0.307
	(0.519)	(0.361)	(0.573)	(0.493)	(0.519)
Any Loan	0.733	0.566	0.716	0.77	0.788
	(0.442)	(0.496)	(0.451)	(0.421)	(0.409)
Years in Ag	10.535	8.798	9.672	10.199	12.507
	(5.514)	(6.438)	(5.4)	(5.081)	(5.026)
N Households	568	71	174	161	162

This table shows summary statistics for agricultural households by township. The table displays means and standard deviations for each variable averaged across household-year observations.

D More on CFE Demands

I provide additional details on the CFE demand system of Ligon (2020) used for the main results. CFE demands satisfy the condition that $\log p_i c_i = a_i(p) + b_i(z) - \beta_i \log \lambda$, where expenditures on good i depend on functions of the price vector p and household characteristics z and are log-linear in λ. β_i is the eponymous constant elasticity, which imposes that the elasticity of expenditure on good i with respect to the marginal utility of expenditure (as opposed to total expenditure) is a constant. This allows for highly non-linear Engel curves and an unrestricted rank of the demand system. Ligon (2020) shows that CFE is the only globally regular demand system in which identical households with different budgets' demands for goods differ only through a common aggregator. The paper also derives an estimator for the MUE that uses disaggregated consumption data. The key assumption for estimation is that observed 0 expenditures can essentially be treated as a missing data problem. While this may appear strong, the assumption essentially requires that welfare can be inferred from observed expenditures and the Frisch elasticities of those goods. See [cite:@ligon2020estimating] and Ligon (2025) for more detail.

What matters for the model in Section 4 is the curvature of utility. The elasticity of λ

Table C3: Summary statistics for agricultural households by township

	All	Chachoengsao	Buriram	Lopburi	Sisaket
Rice	0.691	0.884	0.966	0.007	0.937
	(0.462)	(0.32)	(0.182)	(0.081)	(0.243)
Maize	0.09	0.009	0.004	0.328	0.001
	(0.286)	(0.097)	(0.059)	(0.47)	(0.03)
Farm size	4.797	6.837	2.293	9.663	2.489
	(7.892)	(5.602)	(1.631)	(13.237)	(1.836)
# plots	3.227	3.078	2.097	4.704	3.026
	(2.787)	(2.424)	(1.28)	(4.069)	(1.944)
Any plot rented	0.16	0.395	0.144	0.267	0.025
	(0.367)	(0.489)	(0.351)	(0.443)	(0.155)
Any labor hired	0.682	0.76	0.781	0.849	0.461
	(0.466)	(0.427)	(0.414)	(0.358)	(0.499)
% labor hired	0.287	0.194	0.284	0.539	0.127
	(0.318)	(0.194)	(0.268)	(0.362)	(0.211)
Any fert.	0.89	0.929	0.92	0.803	0.92
	(0.313)	(0.256)	(0.271)	(0.398)	(0.271)
Any equip.	0.907	0.904	0.939	0.923	0.873
	(0.29)	(0.294)	(0.239)	(0.267)	(0.333)
Profit share	0.228	1.056	0.176	0.039	0.172
	(0.688)	(0.905)	(0.564)	(0.606)	(0.585)
N Households	578	73	177	165	163

This table shows summary statistics for agricultural households by township. The table displays means and standard deviations for each variable averaged across household-year observations.

with respect to total consumption is (minus) the coefficient of relative risk aversion. If this elasticity is constant, then CFE reduces to the nested CRRA case.

E Estimation Details

E.0.1 Hedonic land quality

I now describe my approach to distinguish true input distortions, unobserved heterogeneity, and noise. Results in both micro and macro literatures recognize the potential for heterogeneous land quality to bias estimation (Benjamin, 1995; Gollin & Udry, 2021). I address this issue using a hedonic approach. Specifically, I train a model to predict rental values from observed plot features on a random sample of rented plots. These features include area, soil type and quality, histories of drought, flood, erosion, and fertilizer application, proximity

Table C4: Coefficients of variation in factor and output prices by township

	Chachoengsao	Lopburi	Srisaket
Land rent (per rai)	0.5197	0.4376	0.4552
Wage (hourly)	0.7179	0.5652	0.9919
Planting wage (hourly)	0.6822	0.4718	0.8543
Weeding wage (hourly)	0.5899	0.5312	0.5830
Harvest wage (hourly)	0.6151	0.5480	0.9213
Price of rice seed (per kg)	0.2663	0.2069	0.1096
Price of chem. fert. (per kg)	0.1780	0.1413	0.0946
Power tiller rental (per rai)	0.2749	0.4121	0.6040
Large tractor rental (per rai)	0.2093	0.3669	0.2870
Output price of rice (per kg)	0.0944	0.1148	0.0853

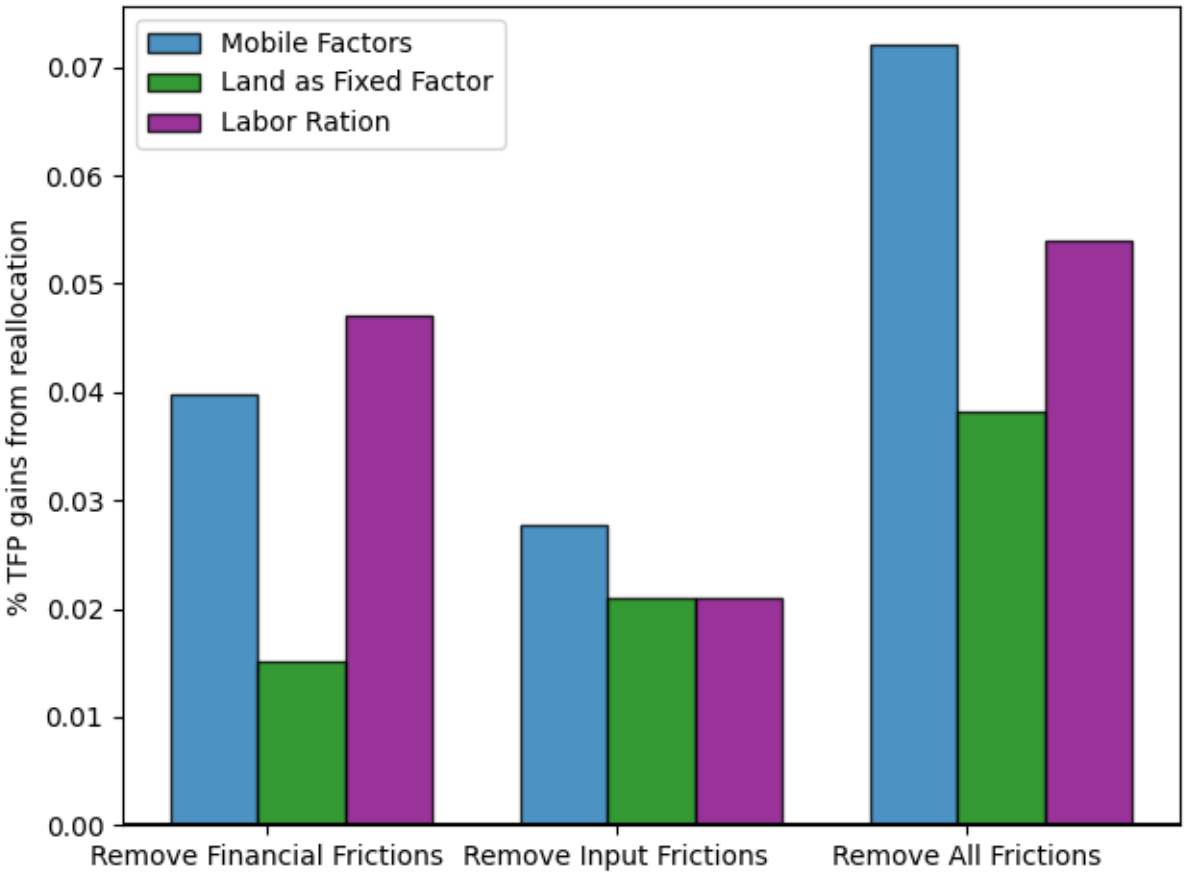
This table shows the coefficients of variation of input and output prices within each township averaged across years. The top panel shows the inputs that I assume are distorted, while the bottom panel shows those that I assume are freely traded. The coefficients of variation are computed at the township-year level after trimming outlier per-unit plot-level expenditures at the upper and lower 2.5% tails and restricting the sample to inputs/outputs with at least 20 observations within a township-year. The three townships shown are those that nearly universally produce rice. The data do not contain the number of days that tractors or power tillers are used — therefore the unit prices I compute are the total expenditure for each type of machinery at the plot level divided by the plot area. Therefore, much of the price dispersion depicted is likely to result from number of days used, machine sizes, or measurement error. Since a more diverse range of crops is grown in Buriram, there is additional heterogeneity due to varieties of seed and fertilizers used for different crops (which I observe). When accounting for this heterogeneity, similar patterns of high price dispersion in land and labor but low price dispersion for traded inputs and outputs emerge.

Table C5: Correlation between financial wedges and financial participation

	Dependent variable:					
	Savings bal.	Debt bal	Credit bal.	Gifts made	Gifts rec'd.	Net gifts
	(1)	(2)	(3)	(4)	(5)	(6)
Λ	15,120.98	37,677.87***	109.22	25,784.96	2,216.36***	−23,568.60
	(11,356.97)	(10,396.81)	(1,314.29)	(17,478.79)	(707.42)	(17,395.17)
Village + Time FE	Yes	Yes	Yes	Yes	Yes	
Observations	4,338	4,338	4,338	4,338	4,338	4,338
Adjusted R^2	0.10	0.06	0.01	0.02	0.16	0.02

Note: *p<0.1; **p<0.05; ***p<0.01

Figure C2: Counterfactual gains from reallocation under input rations



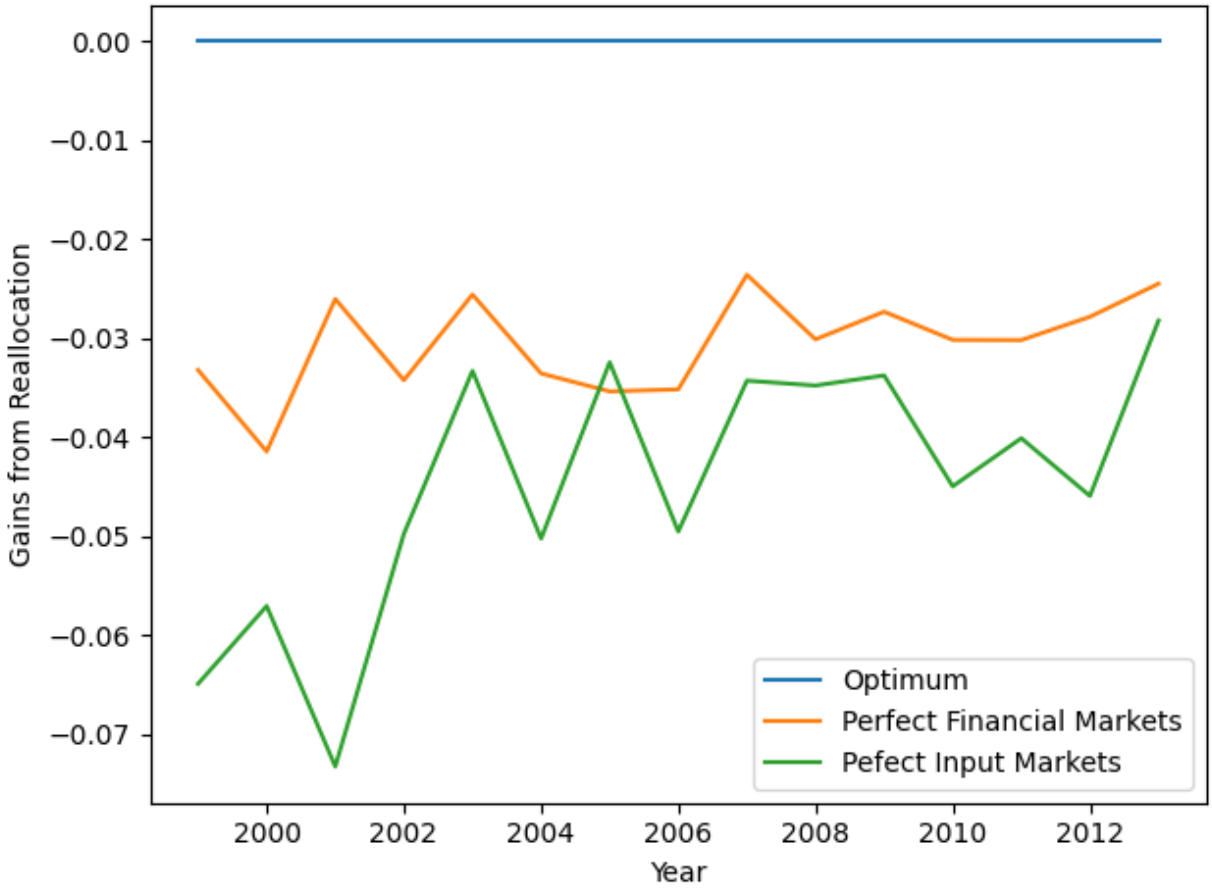
The figure shows the gains from reallocation under the main counterfactuals depending on which factors are mobile within townships. The blue (left) bars reproduce the baseline scenario, in which all factors are mobile and can be reallocated. The green (middle) bars show results holding land fixed at observed levels in all three scenarios, even when relaxing other input frictions. The purple (right) bars show results assuming households with $\tau < 1$ for each labor input face a binding downard ration. Results are computed using CFE demands, fertilizer as the normalizing input for τs, all crops, and aggregating to the farm level.

to water sources, roads, and the household, and (self-reported) sale values.[48] I use cross-validated boosted trees and test the model's fit on a holdout sample, achieving an R^2 of 0.54. I then use the model to assign rental values to plots that were cultivated by the owner, for which no rental price is observed. I then use observed and predicted rental prices as a measure of quality-adjusted land quantities.

There are some caveats to this procedure. First, distorted land markets may not accurately reflect true land quality in prices. While this approach allows for land distortions to take the form of either an implicit tax or a ration, it essentially assumes that there is no distortion to the *relative* prices of observable plot attributes, such as soil and proximity to water sources. Nevertheless, there is no a priori reason to assume that relative prices of

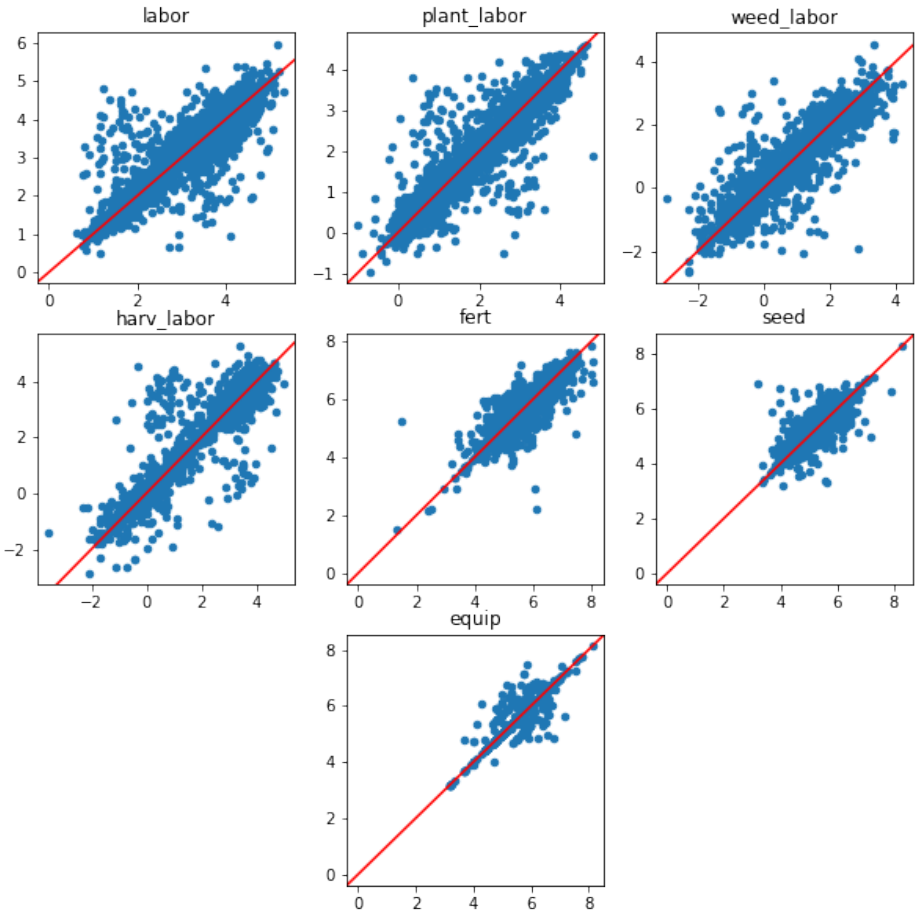
[48] A similar approach is applied by Gordeev and Singh (2023).

Figure C3: Main results over time



This figure plots the main results in Figure 5 disaggregated by year. Results are computed using CFE demands, restricting the sample to rice plots, and aggregating to the farm level. The measure of misallocation is the difference between aggregate TFP under a given allocation and the efficient one, expressed as a percent of modeled TFP. The solid bars compute these using the TFP-based measure of misallocation, using (6). The shaded bars are calculated by taking raw input observed in the data and augmenting them by the estimated τ and Λ, where relevant.

Figure C4: Factor intensities on simultaneous plots

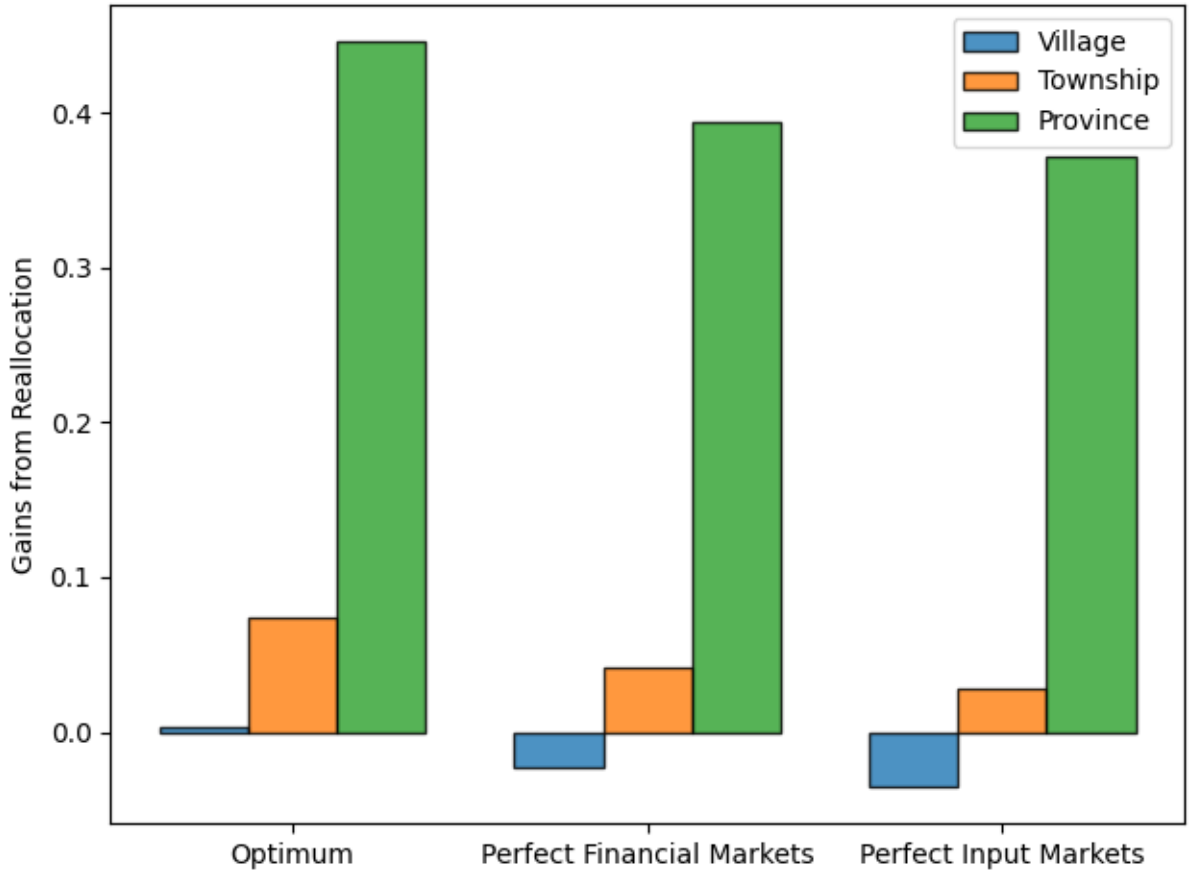


This figure shows scatter plots of the ratio of each input to land on both the largest and smallest rice plots cultivated simultaneously by a household. The 45° line is shown in red — values centered around the 45° suggest that there is no systematic relationship between plot size and input intensity.

Table C6: Production shocks' effect on interhousehold transfers

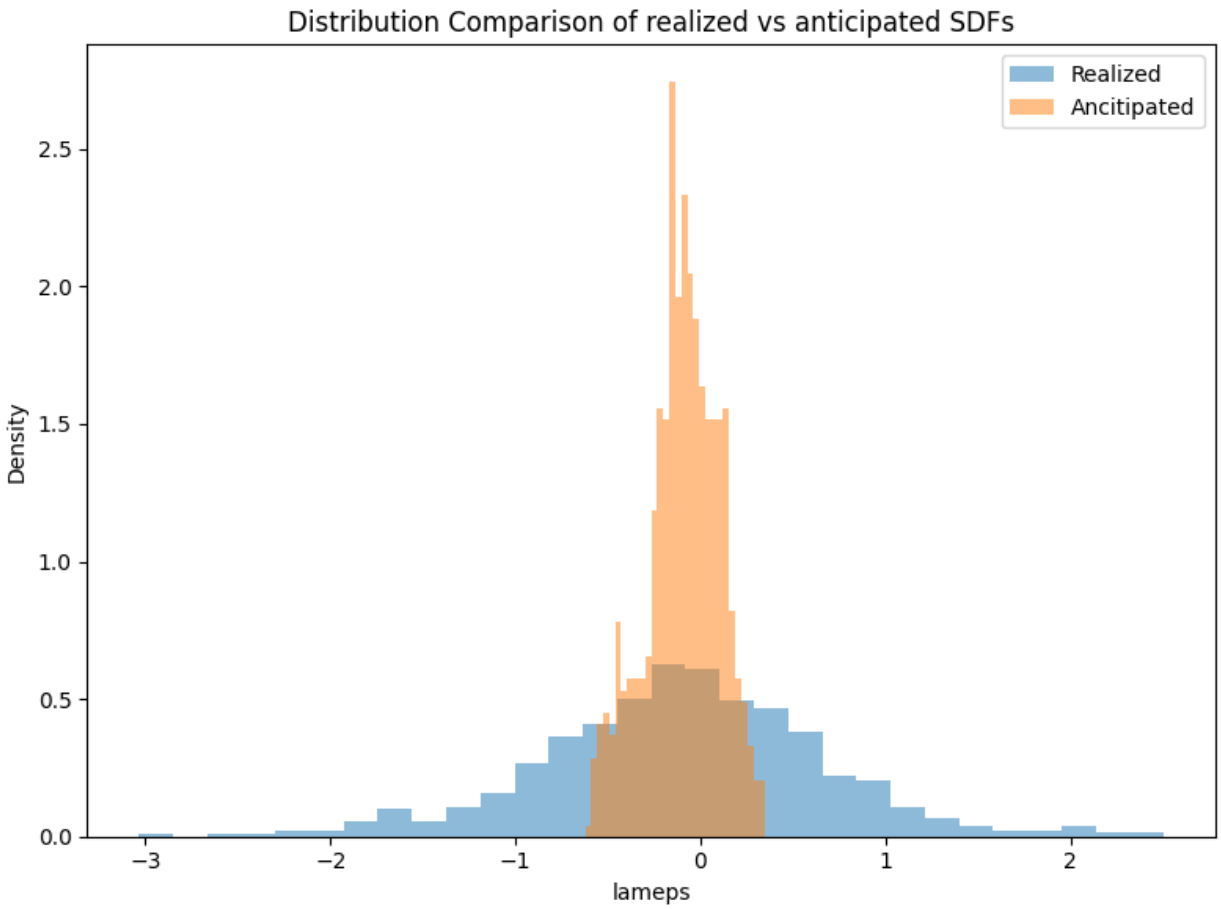
	Dependent variable:				
	Gifts made	log gifts made	Gifts recieved	log gifts recieved	Net gifts
	(1)	(2)	(3)	(4)	(5)
Shock (s.d)	711.71		−146.73*		−858.43
	(662.40)		(86.57)		(707.97)
log shock		0.15**		−0.21***	
		(0.06)		(0.03)	
Village + Time FE	Yes	Yes	Yes	Yes	Yes
Observations	4,338	4,087	4,338	4,322	4,338
Adjusted R^2	0.02	0.14	0.16	0.30	0.02
Note:					*p<0.1; **p<0.05; ***p<0.01

Figure C5: Potential gains from full reallocation



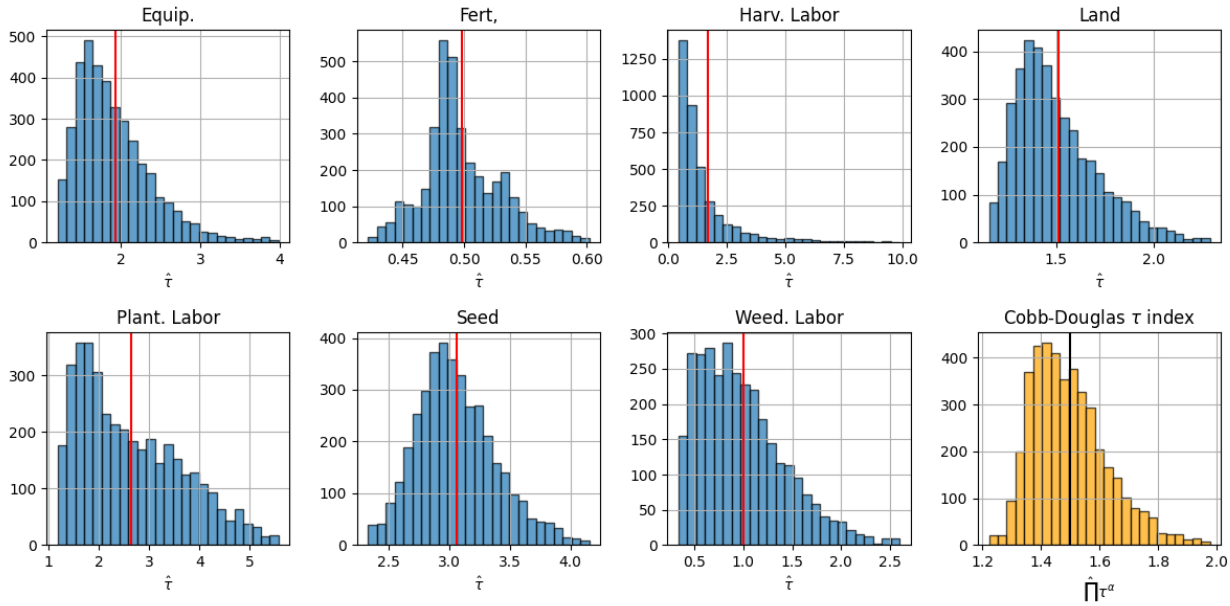
This figure shows the total gains from the efficient allocation as a percent of status quo aggregate TFP when aggregating at the village, township, and national levels.

Figure C6: Distribution of realized and predicted $m\varphi$



This figure plots the distributions of the raw and predicted values of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi$ following the random forest.

Figure C7: Histogram of estimated τ_k and index of τs



This figure plots histograms of the estimated $\hat{\tau}_k$ for each input as well as the Cobb-Douglas price index of τs. The 1% tails are omitted for clarity.

different attributes should be distorted in a particular direction. Another concern is that transacted plots may be selected on unobservable physical attributes. However, the model would capture the value of these attributes toy the extent they are correlated with observable attributes.